MX17 Detector 3 — June 2026 Cosmic-Bench Deep Dive:

micro-TPC angle correlation, drift-velocity measurement, drift-gap determination, and gas diagnosis

Cosmic-bench analysis, Saclay

runs `mx17_det3_saturday_scan_6-27-26` and `mx17_det3_p2_det1_overnight_6-27-26`

2 July 2026 — rev. 4 July 2026 (drift velocity, waveforms, gap & gas: geometry estimator supersedes all time-based fits)

Abstract

We analyse the final det3 weekend campaign: a 2.4 h run at the operating point (resist 490 V, drift 1000 V; configured for 12 h but stopped early for the overnight p2 run), a six-point drift-HV scan (100–1100 V), and two resist-HV scans. Four results: (i) the universally observed *off-diagonal* micro-TPC angle correlation is quantitatively explained as an additive charge-spreading bias, $\tan\theta_{\mathrm{det}} \simeq \tan\theta_{\mathrm{ref}} \pm w/z_{\mathrm{rec}}$ with a spreading width $w \approx 2\,\mathrm{mm}$ — a detector constant, not a rotation and not an analysis bug; (ii) *every* time-based track fit — the production fit and four independent waveform-level estimators — is biased low by 10–20 % through prompt charge sharing between neighbouring resistive strips; the mechanism is demonstrated strip-by-strip in the raw waveforms and closed with a signal-formation simulation, which validates a *geometry* estimator (cluster-extent slope over time-span plateau) as unbiased. The result is $v_{\mathrm{d}}(1000\,\mathrm{V}) = 34.0(15)\,\mu\mathrm{m/ns}$, with 1–3 % statistical precision per 15-minute scan point. Deconvolving the measured sharing from the waveforms ("unsharing") makes the time-based fits agree with geometry and, with one additive calibration constant per plane, yields final micro-TPC angles with 0.19° plateau bias and 1.9° resolution; (iii) the drift gap is mechanically $\approx 30\,\mathrm{mm}$, but only the $\approx 23\,\mathrm{mm}$ of drift column closest to the mesh produces recorded signal: the strip amplitude decays with drift *depth* ($\lambda \approx 16--18\,\mathrm{mm}$, one universal curve across drift fields) — charge from the top of the gap is lost to **attachment** plus threshold in the contaminated gas; (iv) the geometry-based $v_{\mathrm{d}}(E)$ at the 30 mm field scale matches Magboltz for Ar/iso 95/5 + 1.0–1.1 % H_2O (+ ∼1 % air) at RMS $0.84\,\mu\mathrm{m/ns}$, and the measured attachment length matches the accompanying 0.15–0.25 % O_2 — one air/moisture contamination explains both anomalies. Clean gas, mixing errors, and a wrong bottle of Ar/CO_2 90/10 are all excluded (the last by +50 V at iso-gain and zero attachment).

Contents

1 Datasets, alignment and headline performance

1.1 Runs analysed

All data were taken with det3 (`mx17_3`, bulked June 15) in the *top* bench slot: FEU 7 = X strips, FEU 8 = Y strips, nominal $z = 702$ mm, strip pitch 0.78 mm, gas nominally Ar/iC_4H_{10} 95/5 at atmospheric pressure (Saclay, ≈745.8 Torr), non-zero-suppressed DREAM readout (32 samples × 60 ns = 1.92 µs window).

subrun	purpose	dur.	decoded statistics
`long_run_resist_490V_drift_1000V`	operating point	141 min	complete (4 files): 29k events, 47k M3 tracks
`drift_scan_...._drift_{100..1100}V`	drift scan, 6 pts	15 min	∼5k M3 tracks / point
`hv_scan_resist_{425..525}V`	resist scan 1, 11 pts	15 min	∼2k clean rays / point
`hv_scan2_resist_{460..520}V`	resist scan 2, 7 pts	15 min	∼2k clean rays / point
`long_run_p2_det1_sanity_check`	overnight (p2 run)	—	53k rays; current headline det3

Table 1: Weekend det3 datasets. The Saturday long run was configured for 720 min but was stopped after 141 min (trigger-timestamp span; the p2 overnight run started at 01:33 on the same bench) — its 4 file pairs are the complete dataset.

1.2 Alignment

The long run is aligned to the M3 reference tracker with the standard iterative z/rotation/translation procedure (spark veto at 50 strips, clean-cluster subset for the geometry fit). Both weekend runs converge to the *same* geometry — $z = 714$ mm, $\theta = 89.45°$, translation offsets agreeing to 10 µm — confirming the detector did not move between runs. The long-run alignment is therefore used as the seed for every scan point, with only a translation refit per point (observed shifts < 0.1 mm).

1.3 Headline performance

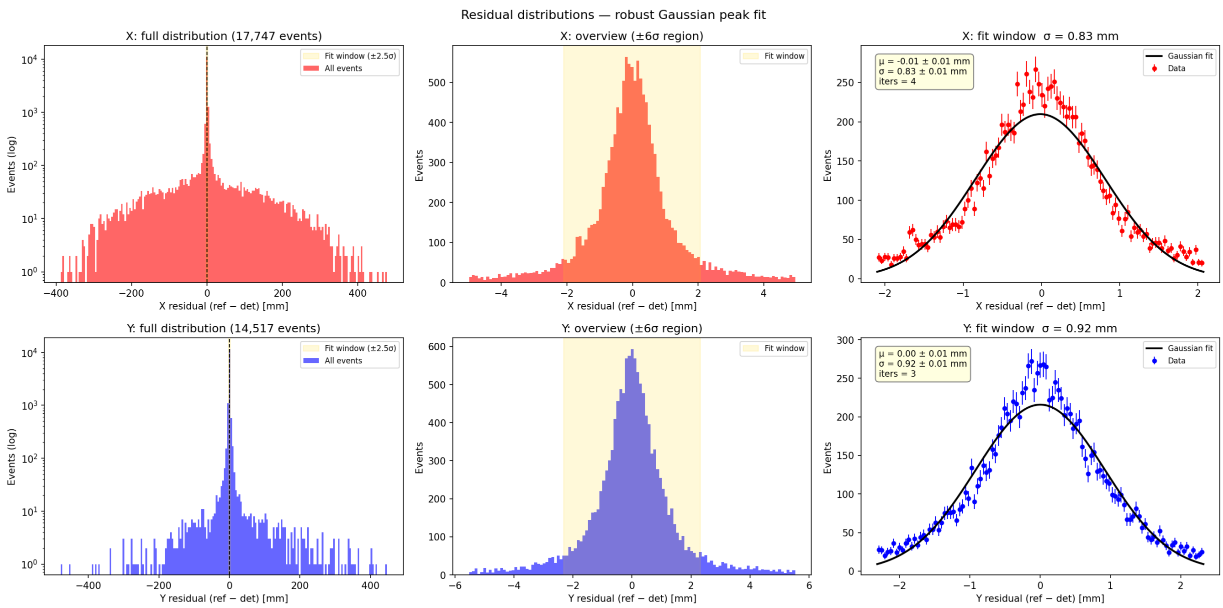

Figure 1: Saturday long run: residuals between the aligned detector hit and the projected M3 track. The Gaussian core gives $\sigma_x = 0.83$ mm, $\sigma_y = 0.92$ mm.

From the efficiency breakdown of the Saturday long run (fixed active box, clean M3 single tracks, $R = 5$ mm matching): the detector fires on 97.6 % of through-going muons, reconstructs an X+Y point for 80.2 %, and 65.8 % land within 5 mm of the track (a 14.4 % *reco-far* tail). The higher-statistics p2 overnight run gives the June-headline 79.7 % within 5 mm with a much smaller tail; the difference (larger far tail in the Saturday run) correlates with the higher spark fraction at resist 490 V (Sec. 8) and can be probed further with the p2 dataset. Either way det3 remains the best detector of the June batch, at sub-mm core resolution in both slots and both weekends.

2 The micro-TPC measurement

2.1 Principle

A single conversion/drift gap of depth g sits above the micromesh; the avalanche charge is read out on strips of pitch $p = 0.78\,\text{mm}$ (Fig. 2). A muon crossing at angle θ (measured from the detector normal, i.e. from the drift direction) leaves an ionisation trail spanning the *full* gap in z and $g \tan\theta$ in x. (In this detector, as Sec. 6 shows, only the fraction of the column within $z_{\text{rec}} \approx 23\,\text{mm}$ of the mesh survives the drift with enough charge to be recorded — everywhere below, "the gap" as seen by the strips means this *visible column* z_{rec}, not the 30 mm electrode spacing.) Electrons drift to the mesh at velocity v_{d}, so the strip at horizontal distance Δx from the track's mesh-crossing point receives its charge later by

$$\Delta t = \frac{\Delta z}{v_{\text{d}}} = \frac{\Delta x}{v_{\text{d}} \tan\theta}. \tag{1}$$

Fitting strip signal time versus strip position (amplitude-weighted straight line, anchored at the earliest hit) yields a slope whose inverse is

$$\frac{\mathrm{d}x}{\mathrm{d}t} = v_{\text{d}} \tan\theta \qquad \Longrightarrow \qquad \tan\theta_{\text{det}} = \frac{1}{v_{\text{d}}}\frac{\mathrm{d}x}{\mathrm{d}t}. \tag{2}$$

Equation (2) is the micro-TPC angle estimator: it needs the drift velocity v_{d} as an external input, and conversely, once an external reference (θ_{ref} from M3) is available, the *correlation between θ_{det} and θ_{ref} measures v_{d}*. That is the core idea exploited throughout this note. In the analysis code the raw fit slope is $s \equiv \mathrm{d}x/\mathrm{d}t$ in µm/ns; the detector-frame slope vector (s_x, s_y) is rotated by the alignment angle ($\theta_{\text{align}} \approx 90°$) into the M3 frame before any comparison, so det-X pairs with ref-Y and vice versa.

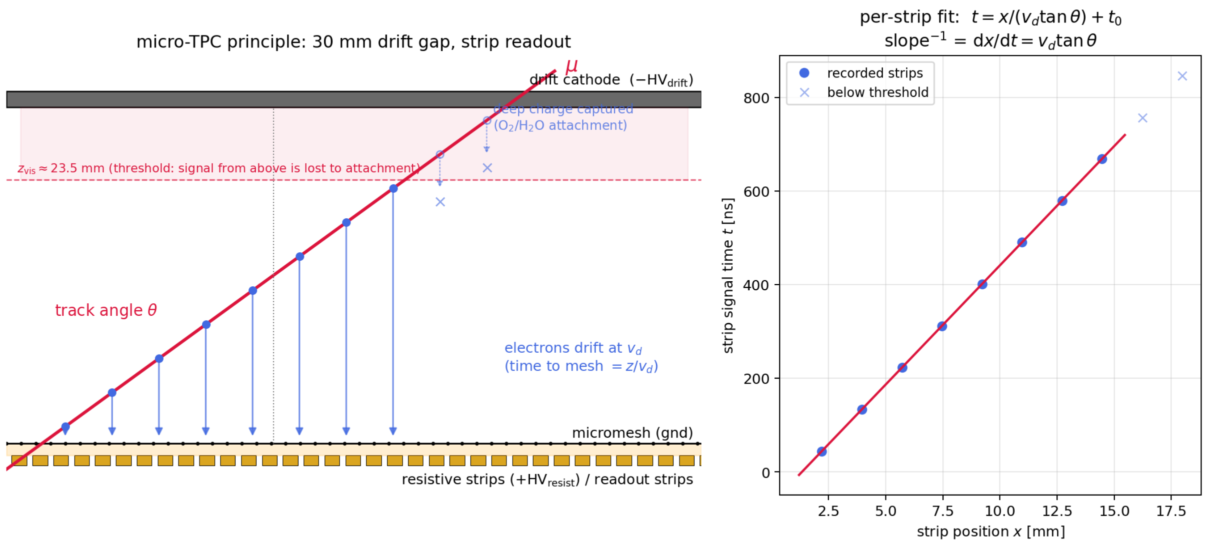

Figure 2: Micro-TPC principle. Left: an inclined muon (red) ionises the gas along the full 30 mm drift gap; electrons (blue) drift down at v_{d} and arrive at the mesh with delays proportional to their production height. Charge from above $z_{\text{rec}} \approx 23\,\text{mm}$ (pink band) is lost to attachment during the drift (Sec. 6). Right: the per-event fit of strip time vs. strip position over the recorded strips; the inverse slope is $v_{\text{d}} \tan\theta$.

2.2 Why the micro-TPC mode matters here

Position resolution (Fig. 1) uses only the charge centroid and is insensitive to everything discussed below. The micro-TPC *slope*, however, is the detector's unique selling point — per-plane track angles from a single thin detector — and it is exactly this quantity that shows the systematic effects analysed next.

3 Why the angle correlation is always off-diagonal

3.1 The observation

Figure 3 shows the production angle-correlation plot for the Saturday long run: reference angle θ_{ref} (M3) versus detector angle θ_{det} (Eq. 2, with v_{d} set by the production diagonal-projection scan). The correlation is strong — the micro-TPC works — but three features stand out, and they appear *identically in every detector of the June campaign* (det2 shown for comparison in Fig. 4):

1. the dense ridge runs *parallel* to the $y = x$ diagonal but is displaced *outward*: $|\theta_{\mathrm{det}}| > |\theta_{\mathrm{ref}}|$ by $\sim$5–6°;
2. there are essentially *no* events with $|\theta_{\mathrm{det}}| \lesssim 5°$ — an exclusion gap around the vertical;
3. events with $\theta_{\mathrm{ref}} \approx 0$ form *horizontal arms* extending to large $|\theta_{\mathrm{det}}|$ of either sign.

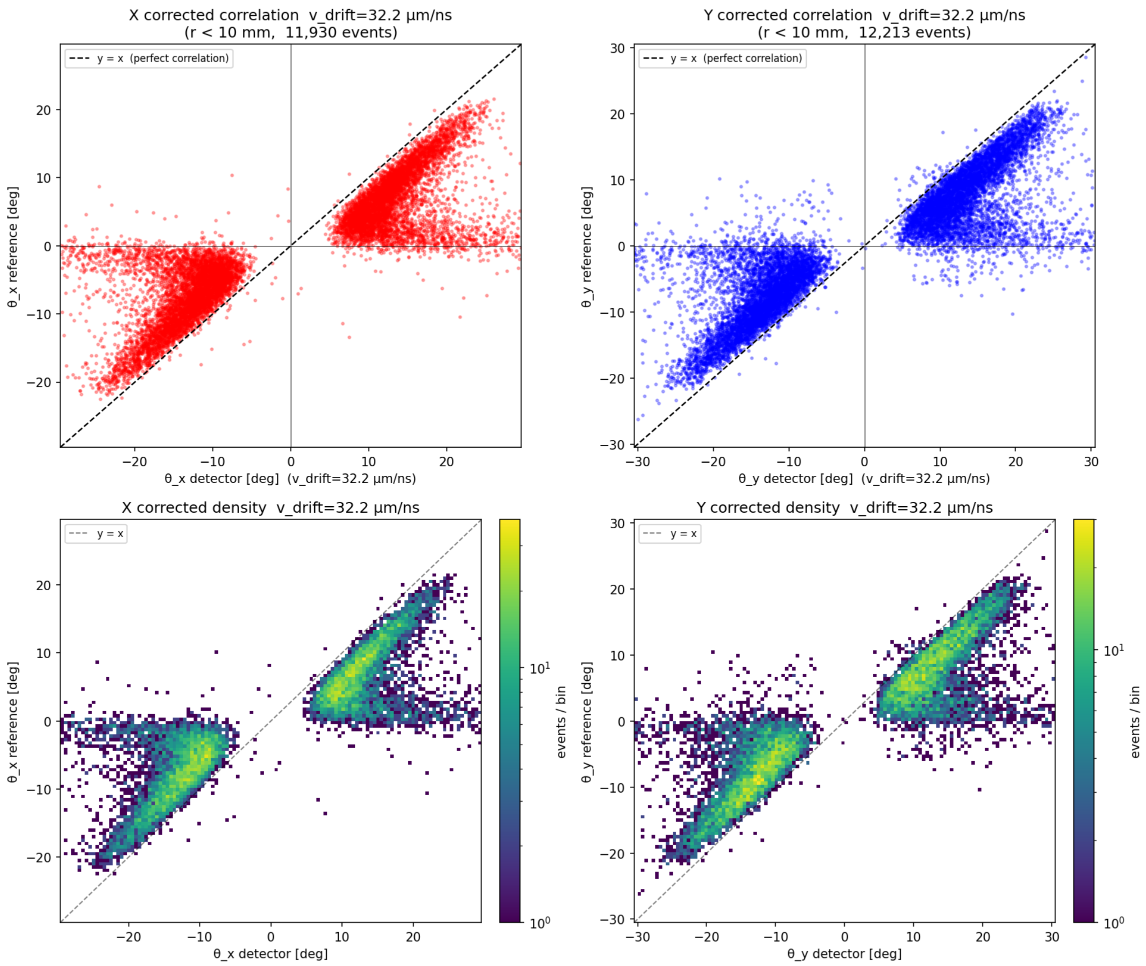

Figure 3: Saturday long run, production angle correlation (top: scatter; bottom: log-density) for the X (left) and Y (right) projections, using the v_{d} from the production σ-scan. The ridge is parallel to, but displaced outward from, the dashed $y = x$ diagonal; note the empty band $|\theta_{\mathrm{det}}| \lesssim 5°$ and the horizontal arms at $\theta_{\mathrm{ref}} \approx 0$.

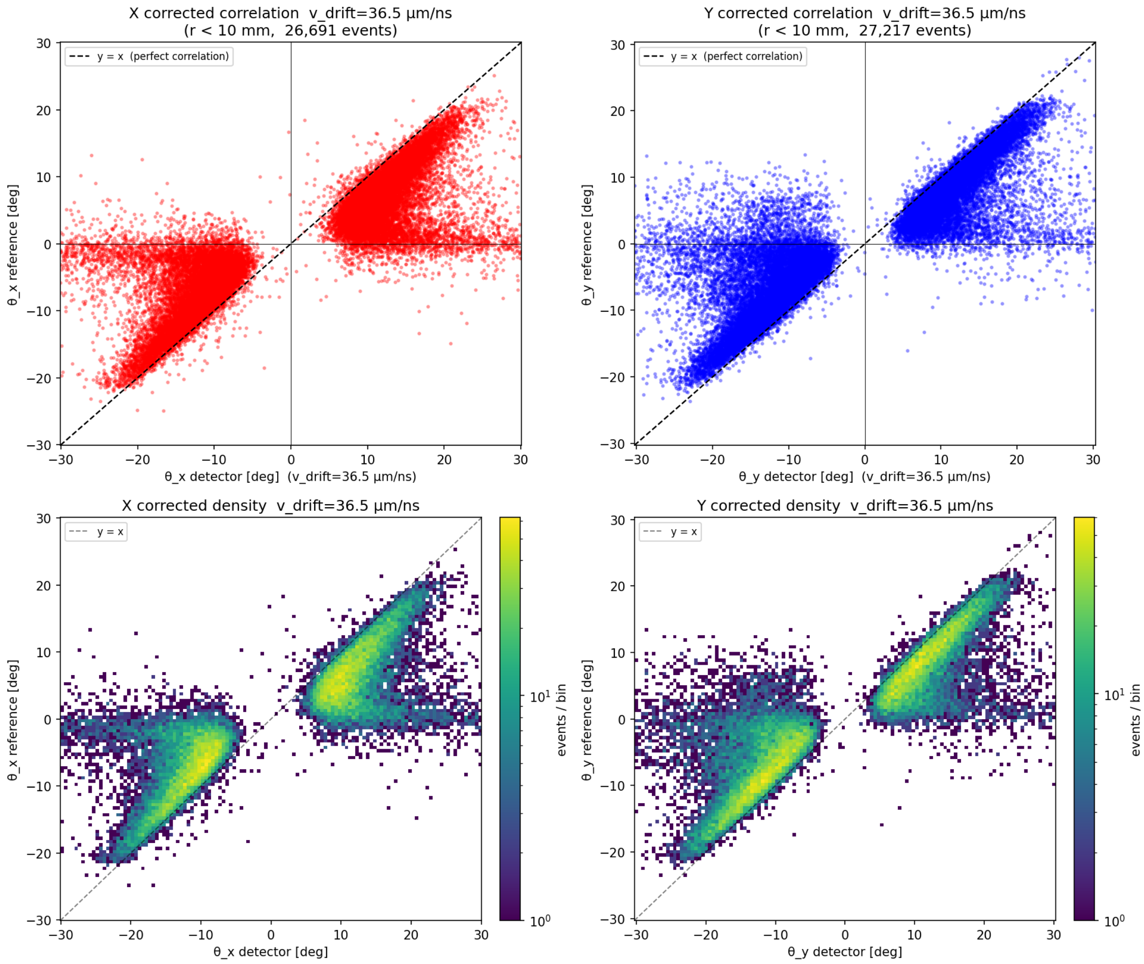

Figure 4: Same plot for *det2* (6-22 overnight run). The off-diagonal structure is identical — this is a property of the detector family, not of one chamber.

3.2 Why it is *not* a rotation or misalignment

A residual in-plane rotation is absorbed by the alignment (the rotation scan converges to 0.05° steps and the slope vector is rotated accordingly). More fundamentally, *any* rotation or constant angular offset would shift the correlation in *one* direction — up or down — whereas the observed displacement **flips sign with the track direction**: tracks leaning one way reconstruct too steep in that direction, tracks leaning the other way too steep in the other. The offset is odd in θ, i.e. it behaves like $\operatorname{sign}(\theta) \cdot \text{const}$, which no rigid-body transformation produces. An out-of-plane tilt of the detector would also be a one-directional shift, and is excluded for the same reason.

3.3 The mechanism: fixed time span, inflated cluster width

The key geometric fact (Fig. 5): **every cosmic crosses the full drift gap**, so the recorded cluster always spans the entire *visible* column depth z_{rec} in z (the fixed depth at which drifting charge drops below threshold; Sec. 6). Consequently the *time* extent of every cluster is the same,

$$\Delta t_{\text{cluster}} \approx T_{\text{sat}} = z_{\text{rec}}/v_{\text{d}}, \qquad \text{independent of } \theta, \tag{3}$$

while the *spatial* extent is

$$\Delta x_{\text{cluster}} \approx \underbrace{z_{\text{rec}} \tan\theta}_{\text{geometric}} + \underbrace{w}_{\text{non-geometric}}, \tag{4}$$

where w collects everything that widens the charge footprint on the strips without carrying drift-time information: charge spreading across the resistive layer, transverse diffusion, capacitive coupling to neighbours, and δ-electrons. The fitted slope is then, schematically (writing $z_r \equiv z_{rec}$),

$$\frac{dx}{dt} \approx \frac{z_r \tan\theta + w\,\mathrm{sign}(\theta)}{z_r/v_d} = v_d\left(\tan\theta \pm \frac{w}{z_r}\right) \qquad \Longrightarrow \qquad \boxed{\tan\theta_{det} \approx \tan\theta_{ref} \pm \frac{w}{z_{rec}}} \tag{5}$$

an **additive, angle-independent excess in tan-space** whose sign follows the track sign. (The sign attaches to w because the fit interprets the extra width as extra track length in the direction the track already leans; for a perfectly vertical track the sign is random — hence the horizontal arms.)

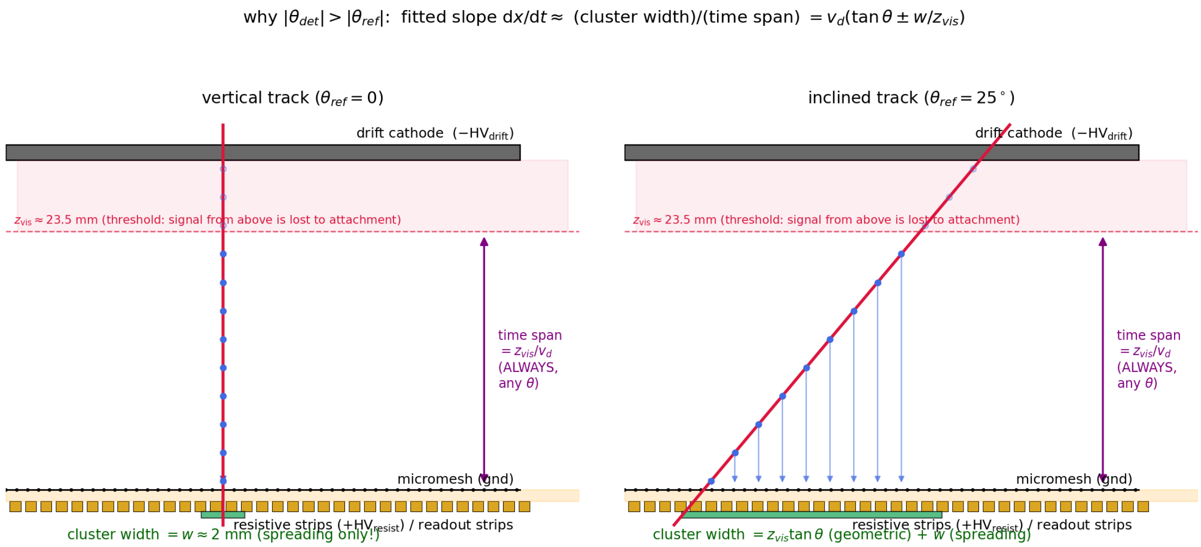

Figure 5: The off-diagonal mechanism. Left: a vertical track — the time span still covers the full visible column (z_{rec}/v_d), but the cluster width is pure spreading, $w \approx 2$ mm; the fit converts w into a fake angle $\arctan(w/z_{rec}) \approx 6.5°$ of random sign. Right: an inclined track — the width gains the geometric part $z_{rec}\tan\theta$, but w still adds on top, always *increasing* $|\theta_{det}|$.

This single mechanism explains all three observations at once:

- **parallel outward ridge**: Eq. (5) is a constant displacement w/z_{rec} in tan-space; converted to degrees it is a nearly constant $\sim \arctan(w/z_{rec})$ shift over the populated range;
- **exclusion gap**: no cluster can fit a slope smaller than $\sim w/z_{rec}$, because even a vertical track's cluster is w wide over a full-column time span — $|\theta_{det}| \gtrsim \arctan(w/z_{rec})$ always;
- **horizontal arms**: for $\theta_{ref} \approx 0$ the geometric width vanishes, the sign of the fitted slope is set by noise, and the magnitude is again $\sim w/z_{rec}$ or larger (the events that pass the $\geq$ 4-strip cut at small angle are precisely those with the largest spreading).

And because w (a resistive-layer/diffusion property) and z_{rec} (set by the shared gas supply and thresholds, Sec. 6) are common to the whole detector family, **every chamber shows the same off-diagonal pattern** — exactly what is observed.

3.4 Quantitative verification

All of this is testable in the data of the 2.4 h run (`13_tpc_angle_bias.py`); Figs. 6 and 7. The ratios in this subsection are quoted in the production time-fit frame ($v_d = 28.1\,\mu$m/ns, used to make these plots); the ~20 % velocity-scale correction of Secs. 4–5 rescales $\tan\theta_{det}$ and w/z_{rec} coherently and changes neither the mechanism nor the spreading width w:

- **the excess is flat in tan-space**: the median of $|\tan\theta_{\rm det}|-|\tan\theta_{\rm ref}|$ is constant at $w/z_{\rm rec} = 0.115$ for $|\theta_{\rm ref}| \gtrsim 6°$, in both planes (Fig. 6, bottom row). $\arctan(0.115) = 6.6°$ — the observed ridge displacement;
- **wider clusters carry more bias**: splitting by cluster size (4–6 / 7–9 / ≥10 strips) orders the excess exactly as the spreading picture predicts;
- **the time span saturates**: the median cluster time span rises to a plateau $T_{\rm sat} \approx 690$ ns for $|\theta_{\rm ref}| > 10°$ — the full-gap drift time (Fig. 7, top row). (Its decrease towards $\theta = 0$ is a per-strip integration effect: a vertical track's charge lands on few strips whose fitted times average over the column, shrinking the strip-to-strip span.);
- **the spatial width has a floor**: the median cluster extent at $\theta_{\rm ref} \approx 0$ is 3–5 mm where geometry predicts ≈ 0; at larger angles it grows parallel to the geometric expectation $z_{\rm rec}\tan\theta$, offset by the same floor (Fig. 7, bottom row).

The ridge-fit intercepts (Sec. 4) give the same number from a completely different projection of the data: $|b|/v = w/z_{\rm rec} = 0.115$, i.e. a spreading width $w = |b| \cdot T_{\rm sat} = 2.2$ mm (≈3 strips), a velocity-independent number. Finally, overlaying the parameter-free prediction of Eq. (5) on the measured density (Fig. 8) tracks the ridge in both planes.

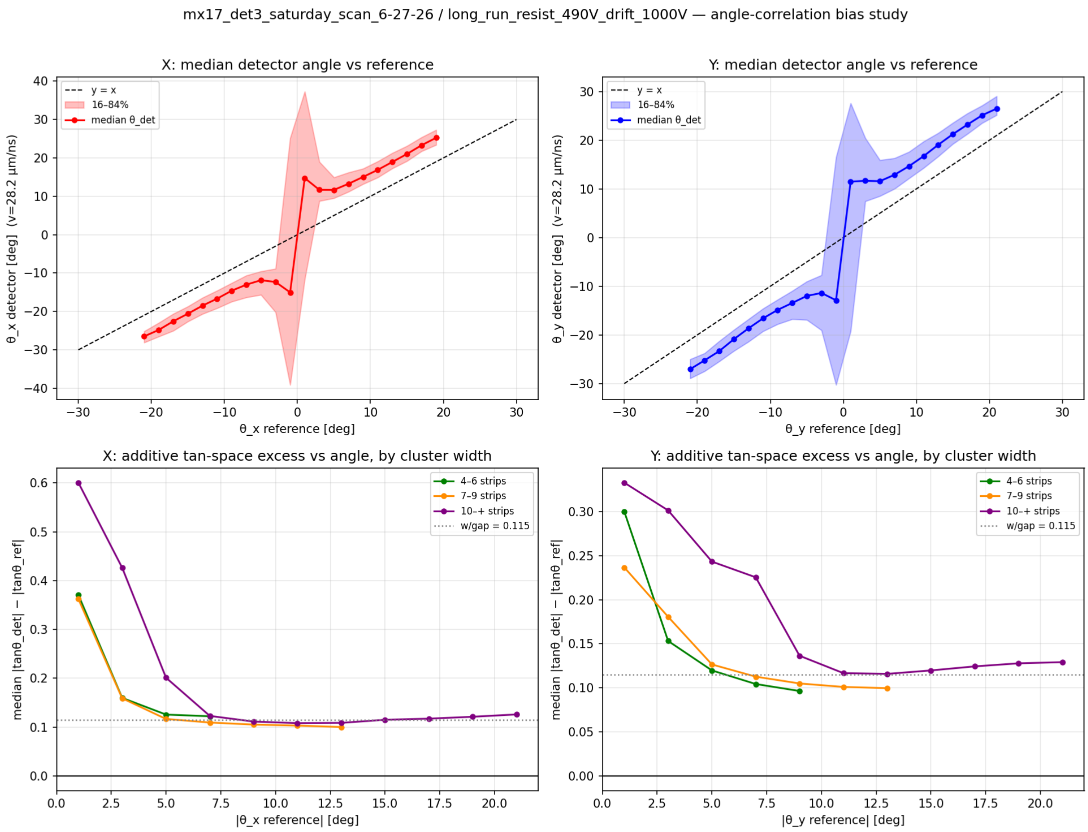

Figure 6: Bias study on the 2.4 h run. Top: median $\theta_{\rm det}$ vs $\theta_{\rm ref}$ — parallel to the diagonal, displaced outward, with the sign flip through zero. Bottom: the additive tan-space excess $|\tan\theta_{\rm det}|-|\tan\theta_{\rm ref}|$ vs $|\theta_{\rm ref}|$, split by cluster size; it is flat at $w/z_{\rm rec} = 0.115$ (dotted) above $\sim 6°$ and grows for wide clusters. The rise at small angles is the ≥ 4-strip selection (only heavily-spread clusters survive).

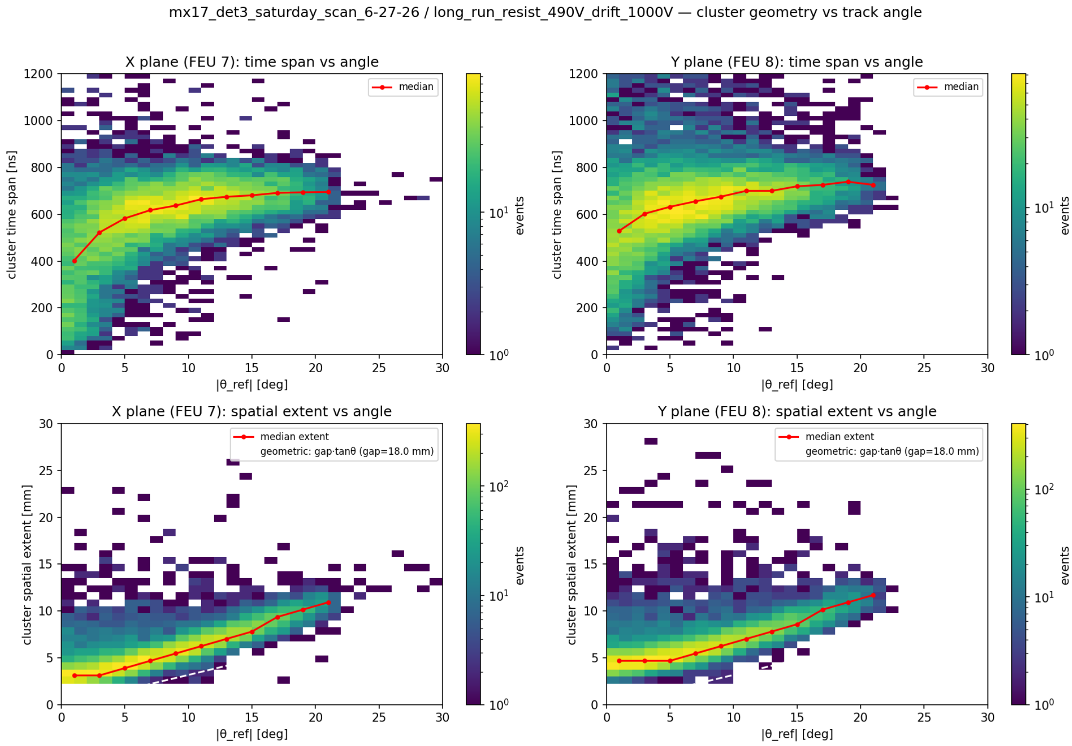

Figure 7: Cluster geometry vs track angle (2.4 h run). Top: cluster *time* span — saturates at $T_{\text{sat}} \approx 690$ ns (the drift time across the visible column) for inclined tracks. Bottom: cluster *spatial* extent — a 3–5 mm non-geometric floor at $\theta = 0$, then growth parallel to the geometric $z_{\text{rec}} \tan\theta$ (white dashed).

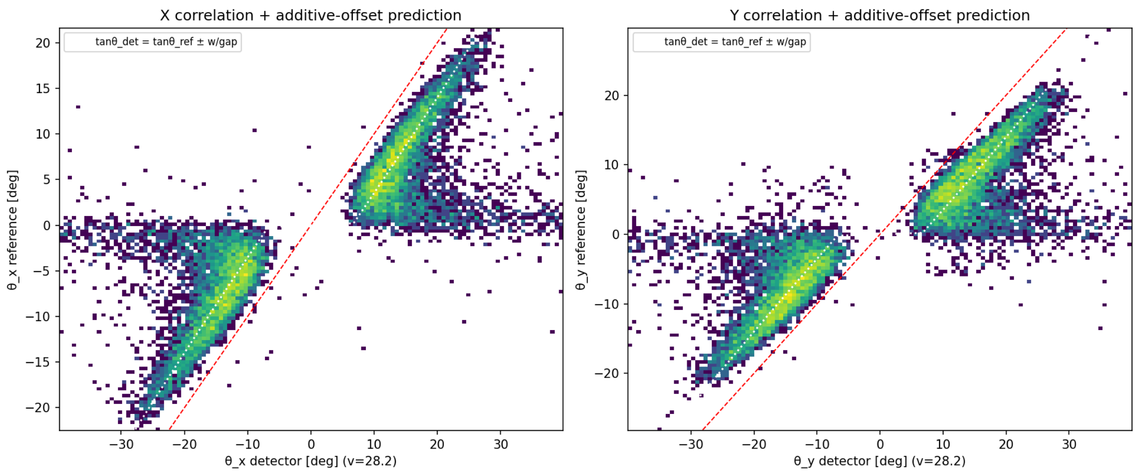

Figure 8: Measured correlation density with the parameter-free prediction of Eq. (5), $\tan\theta_{\text{det}} = \tan\theta_{\text{ref}} \pm w/z_{\text{rec}}$ (white dotted), using the independently fitted $w/z_{\text{rec}} = 0.115$. The red dashed line is $y = x$.

3.5 Consequences

1. **The production σ-scan drift velocity is biased.** The production estimator (Fig. 9) scans v_{d} and minimises the width of $|\theta_{\text{ref}}| - |\theta_{\text{det}}|$; the additive offset pulls its minimum high ($v_{\text{d}} = 30.5/34.0$ µm/ns for X/Y on this run versus the ridge 28.1 µm/ns — itself biased low by the charge-sharing mechanism of Sec. 5; the physics number is the geometric 34 µm/ns of Sec. 4). Its $|\cdot|$-folding also mixes the sign-following offset into the width. Use the geometry

estimator for velocity.

2. **A first-order angle correction exists**: subtract $\mathrm{sign}(\theta_{\mathrm{det}})\, w/z_{\mathrm{rec}}$ in tan-space. It sharpens inclined tracks but cannot recover $|\theta| \lesssim 5°$, where the angular information is genuinely destroyed by spreading.

3. **Angular resolution** (after slicing in θ_{ref}, i.e. with the local bias absorbed): $\sigma_\theta \approx 1.5 - -2°$ per plane for $|\theta| \gtrsim 5°$, degrading to $\sim 6°$ near normal incidence (Fig. 10).

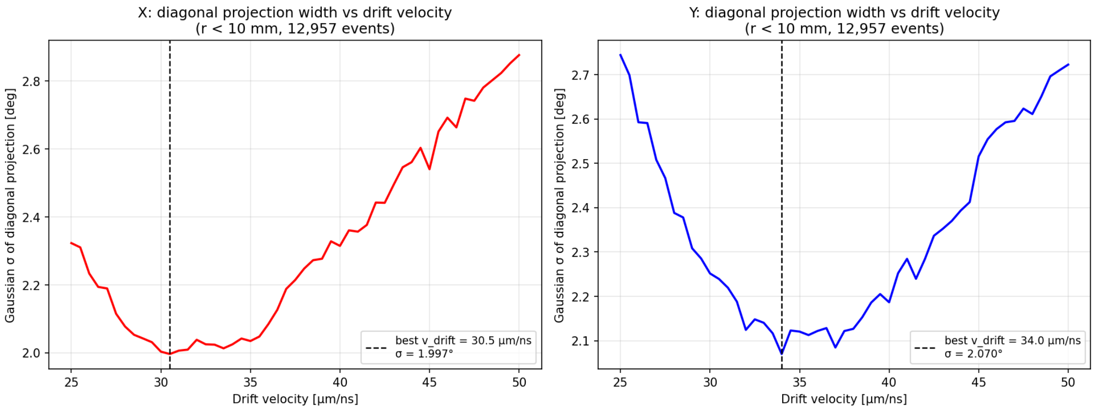

Figure 9: Production diagonal-projection σ-scan on the Saturday run. The minima (X: 30.5, Y: 34.0 µm/ns) are biased high by the additive offset; kept here for reference only.

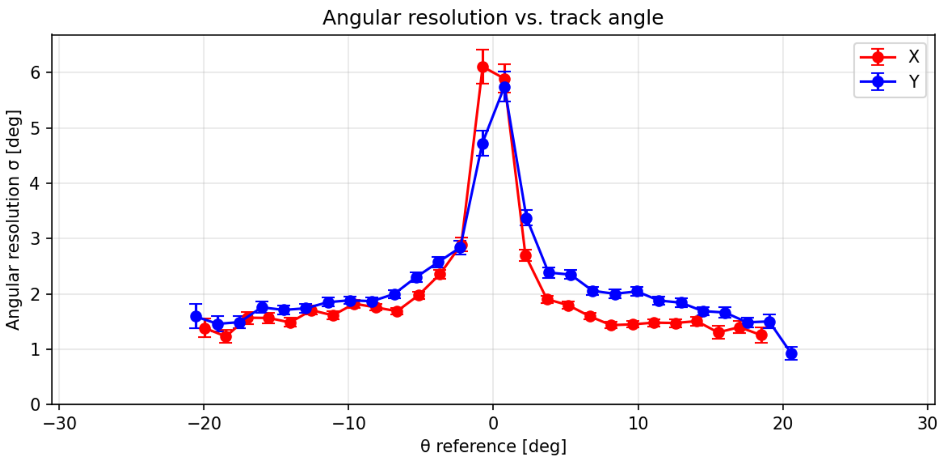

Figure 10: Micro-TPC angular resolution vs track angle (2.4 h run): $\sigma_\theta \approx 1.5 - -2°$ per plane for inclined tracks; the method collapses below $\sim 5°$ as predicted by the spreading mechanism.

4 Drift-velocity measurement from the drift scan

4.1 First attempt: the ridge fit — and its hidden assumption

Rewriting Eq. (5) in terms of the *raw fitted slope* $S \equiv v_{\mathrm{d}} \tan\theta_{\mathrm{det}}$ (in µm/ns, after rotation to the M3 frame):

$$S \;=\; v_{\mathrm{d}}\, \tan\theta_{\mathrm{ref}} \;\pm\; v_{\mathrm{d}}\, \frac{w}{z_{\mathrm{rec}}}. \tag{6}$$

If the spreading term is angle-independent, this is a straight line whose slope is v_{d} and whose intercept absorbs the offset. Per-track-sign straight-line fits (iteratively 3σ-clipped, window $0.06 <$

$|\tan\theta_{\mathrm{ref}}| < 0.55$; four independent X/Y $\times$ sign fits combined) give the *ridge* velocities of Table 2 — at 1000 V, 28.1(7) µm/ns.

The hidden assumption is that w does not depend on angle. It does: the ridge is measurably *convex* — refitting in split windows gives 26–27 µm/ns for $|\tan\theta_{\mathrm{ref}}| < 0.28$ and 31–32 µm/ns for $|\tan\theta_{\mathrm{ref}}| > 0.28$, in both planes, with the intercept simultaneously shrinking (Fig. 11). Any angle dependence of the spreading trades one-for-one into the fitted slope, and a "flat residual" check cannot detect it (a linear $w(\tan\theta)$ is mathematically degenerate with a change of v_{d}). The ridge numbers are therefore kept only as the *time-fit* reference; Sec. 5 shows from the raw waveforms why every time-based fit shares this bias.

4.2 The geometry estimator

The degeneracy is broken by an observable with *no time axis at all*: the cluster spatial extent (strip count $\times$ pitch) versus the M3 track angle. Its slope is the recorded column depth,

$$\frac{\mathrm{d\,extent}}{\mathrm{d}\tan\theta_{\mathrm{ref}}} = z_{\mathrm{rec}} + w'(\theta), \tag{7}$$

with the spreading floor confined to the intercept, and the drift time across the *same* recorded column is the time-span plateau T_{sat}. Their ratio is the drift velocity:

$$\boxed{v_{\mathrm{d}}^{\mathrm{geom}} = \frac{\mathrm{d\,extent}/\mathrm{d}\tan\theta_{\mathrm{ref}}}{T_{\mathrm{sat}}}} \tag{8}$$

Both planes give identical extent slopes (23.2 mm per unit tan on the 2.4 h run, Fig. 11), beautifully linear over the populated range, and the estimator is stable when both axes are restricted to the amplitude *core* of the clusters ($\geq$20/30/40 % of the cluster maximum give $v_{\mathrm{d}} = 35.0 \pm 0.6$ at 1000 V; `23_core_geometry_vdrift.py`). The signal-formation closure test of Sec. 5 confirms this estimator recovers the true velocity within $\sim$3 % under data-like conditions, while all time-fit estimators are biased low.

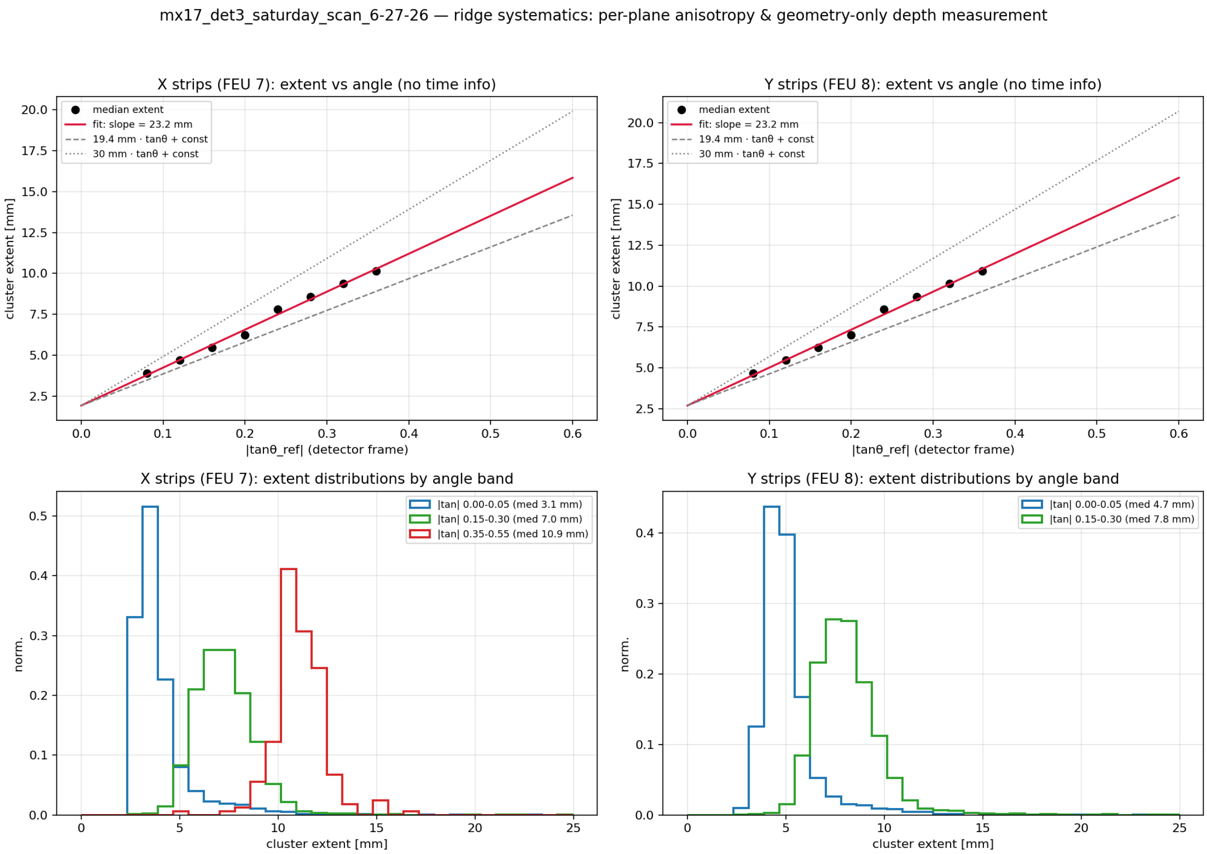

Figure 11: The geometry measurement (`20_ridge_systematics.py`). Top: median cluster extent [mm] vs $|\tan\theta_{\mathrm{ref}}|$ (detector frame) — both planes give a slope of 23.2 mm per unit tan, between the 19.4 mm implied by the (biased) ridge fit and the 30 mm mechanical gap. Bottom: extent distributions by angle band; the $\theta \approx 0$ floor is 3.1 mm (X) / 4.7 mm (Y) — the anisotropy expected from resistive strips running along y.

4.3 Results

drift HV [V]	E [V/cm]	v_d ridge (biased) [µm/ns]	v_d **geometry** [µm/ns]	T_sat [ns]	note
100	33	—	—	—	drift time ≫ window
300	100	5.70 ± 0.54	—	(1198)	window kills extent lever arm
500	167	12.77 ± 0.43	12.37 ± 0.39	(1361)	window-truncated column
700	233	19.45 ± 0.53	21.61 ± 0.53	992	
900	300	26.17 ± 0.84	30.03 ± 0.64	754	
1000	333	28.12 ± 0.65	33.91 ± 0.25	690	**2.4 h long run**
1100	367	29.44 ± 0.47	35.13 ± 0.93	661	

Table 2: Drift-velocity measurements (`14_drift_velocity_scan.py`, `21_geometry_vdrift_scan.py`); $E = V/3.0\,\mathrm{cm}$. The geometry estimator is the physics result; the ridge column is retained to document the time-fit bias, which grows from ≈0 at 500 V to ≈20 % at 1000 V. Absolute scale systematic on $v_\mathrm{d}^{\mathrm{geom}}$: ∼5 % (residual spreading term w', bounded by the core-fraction stability and the toy closure).

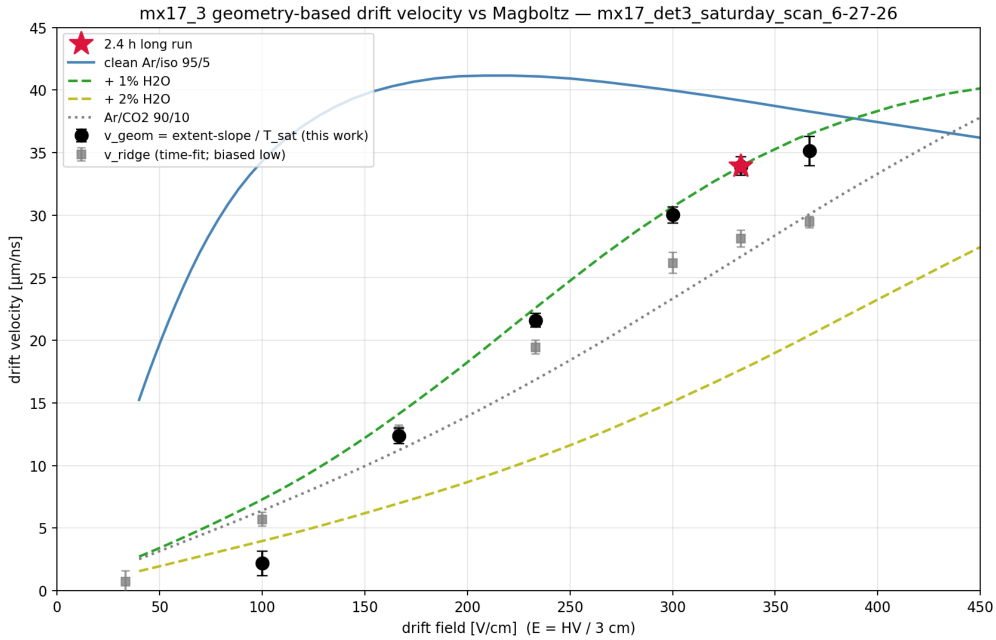

Figure 12: Geometry-based $v_\mathrm{d}(E)$ (black; star = 2.4 h run) against the biased ridge values (grey squares) and Magboltz predictions. The measured points thread the Ar/iso 95/5 + 1 % H_2O curve (Sec. 7).

Key points:

- **Statistics are not a limitation**: 15-minute scan points give 1–3 % statistical errors.
- **The recorded column is constant**: extent slopes lock at 21–23.5 mm for $\geq$ 700 V (Sec. 6).
- **The low-HV points die by window truncation**: at 100–300 V most charge arrives after the 1.92 µs DREAM window closes; the same truncation (plus stronger low-field attachment) drives the efficiency turn-on of the drift scan (Fig. 13): 44 % at 100 V rising to 83 % at $\geq$ 700 V — the quantitative version of "det3 needs drift $\geq$ 700 V".

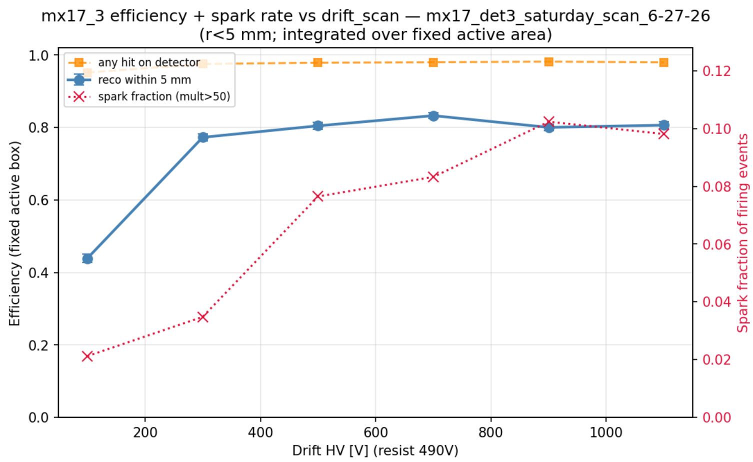

Figure 13: Reconstruction efficiency vs drift HV (drift scan, fixed active box). The turn-on tracks the fraction of the drift column whose charge arrives inside the readout window (and survives attachment); the plateau begins once the surviving column's drift time fits in the window.

5 Why every time-based fit is biased: waveforms, charge sharing, and a simulation closure

The 20 % gap between the ridge and geometry velocities demands an explanation at the signal level. The decoded (non-zero-suppressed) DREAM data of the long run — 512 channels × 32 samples per event and FEU — were pulled from the lxplus archive and analysed strip by strip (`24_waveform_investigation.py`).

5.1 What the waveforms show

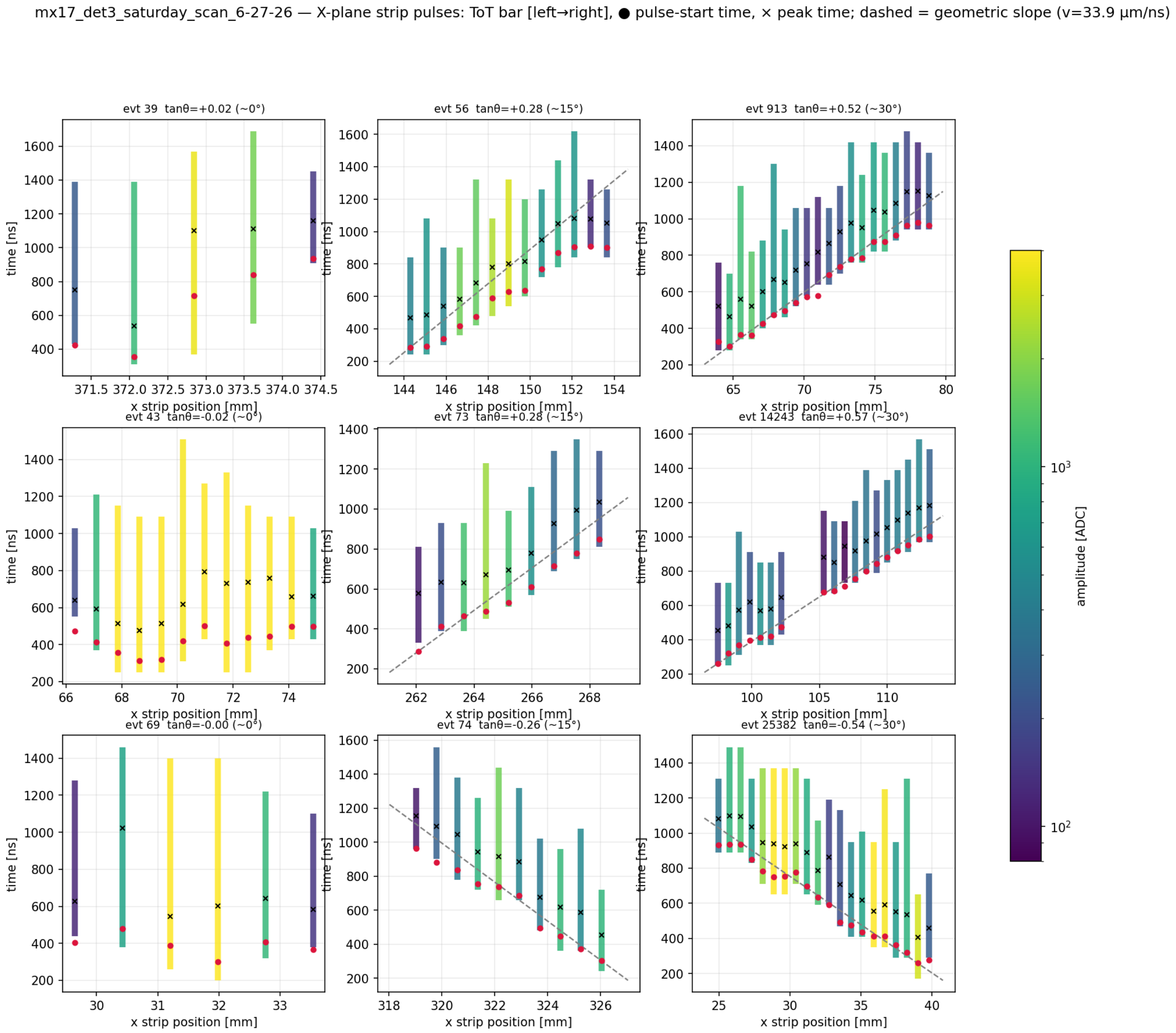

Figure 14: Per-strip pulse structure for reference tracks at ∼0/15/30° (columns): time-over-threshold bars vs strip position, colour = amplitude; • = pulse-start time, × = peak time; dashed = the geometric slope at $v_\mathrm{d}^\mathrm{geom}$. The time ladder follows geometry through the cluster core but *flattens at both ends*: low-amplitude mesh-end strips start late, deep-end strips fire early.

Three facts, from Figs. 14 and 15:

1. **The hit times are faithful to the waveforms.** Re-deriving each strip's time from the raw samples with four independent estimators — fixed threshold, constant-fraction (50 %), rising-edge fit, derivative maximum — and refitting the track slope on core strips gives v_d = 28.0–30.5 µm/ns: the same as the production pulse fit. The bias is *not* in the pulse fitting.

2. **The pulse shapes are healthy**: the rise scale ($t_{\text{peak}} - t_{\text{CFD}}$) is $\approx$ 140 ns for every amplitude class (the shaper), and core-strip time residuals from the track line show no amplitude walk. Skirt strips carry late tails of 200–400 ns.

3. **The ladder, not the pulses, is distorted**: within one cluster the time–position relation is S-shaped — geometric in the middle, flattened at both ends — so a straight-line fit returns a slope steeper (in ns/mm) than geometry, i.e. a velocity that is too low, while the endpoints span less time than the geometric column implies.

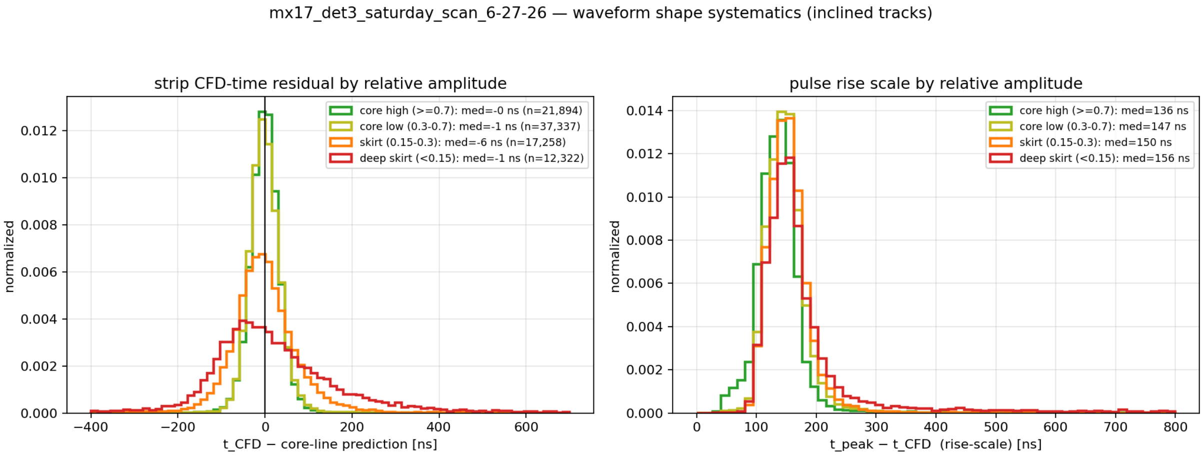

Figure 15: Waveform shape systematics (inclined tracks). Left: CFD-time residual from the core line by relative amplitude — core strips are tight and unbiased; skirt strips have heavy late tails. Right: rise scale is $\approx$ 140 ns for all classes — pulse shapes are not the problem.

5.2 The mechanism: prompt charge sharing

On a resistive-strip anode a substantial fraction of every strip's charge appears (almost) promptly on its neighbours — capacitive coupling plus dispersion; the along-y RC component additionally widens the y-measurement footprint (the 4.7 mm vs 3.1 mm floors of Fig. 11). Figure 16 illustrates the chain of consequences. For an inclined track each strip's waveform is a superposition of its own (direct) charge and *earlier* charge shared from its shallower neighbour (panel b): the pulse start is pulled early or late depending on the local amplitude balance. Where direct charge is strong (core middle) the distortion is small; at the deep end, where attachment has weakened the direct charge, the shared component dominates and the strip fires at nearly its neighbour's time. The time–position ladder is therefore S-shaped (panel c): geometric in the middle, flattened at both ends — and any straight-line fit through it returns a slope that is too steep in ns/mm, i.e. a velocity that is too low. Note the trap: because *all* per-strip times inherit this distortion, re-deriving them from the raw waveforms with better pulse estimators changes nothing — the bias is in the physics of the signal, not in the fitting.

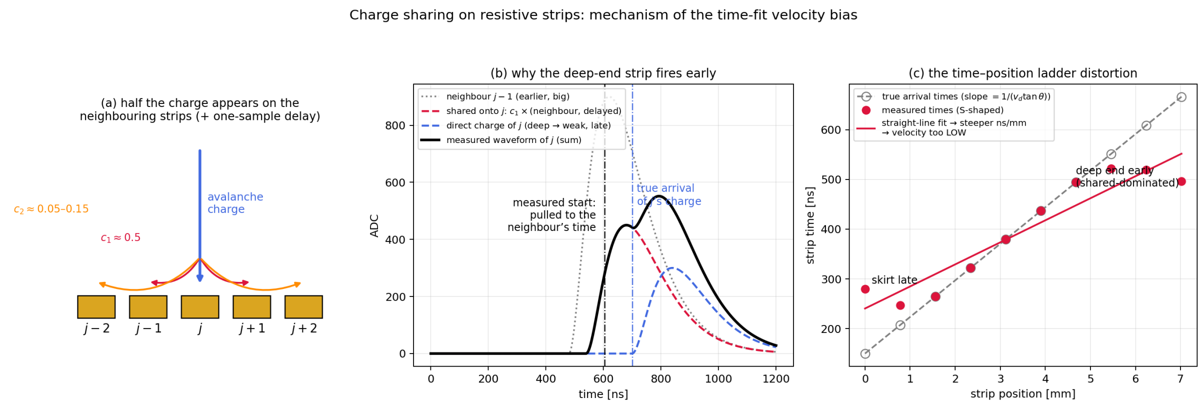

Figure 16: The charge-sharing mechanism, schematically. (a) The avalanche charge of one strip appears at $c_1 \approx 0.5$ on each first neighbour and $c_2 \approx 0.05$–0.15 on each second neighbour, with a one-sample (≈ 60 ns) delay. (b) A weak (deep-end) strip's measured waveform is dominated by the shared copy of its earlier neighbour; its fitted start time is pulled to the neighbour's time. (c) The resulting S-shaped time–position ladder: a straight-line fit is steeper (ns/mm) than the true drift relation, biasing the velocity low.

5.3 Closure test with a signal-formation toy MC

A toy simulation (`25_signal_formation_toy.py`) generates waveform-level events with known v_d^{true}: ionisation clusters along the track, exponential gain fluctuations, attachment, transverse and longitudinal diffusion, prompt charge sharing to $\pm1/\pm2$ neighbours, CR–RC^2 shaping (peak 140 ns), 60 ns sampling, noise and thresholds — then runs the *same* estimators as on data. With sharing fractions of $\sim$35 %/12 % and $\lambda_{\text{att}} \approx 10$ mm the toy reproduces the data pattern simultaneously: extent slope 23.6 mm (data 23.2), $T_{\text{sat}} = 695$ ns (data 690), extent floor 4.7 mm, and time-fit velocities 15–25 % below truth. Scanning v_d^{true} at fixed parameters:

- the **geometry estimator recovers v_d^{true} within $\sim$3 %** at every velocity tested (28–37 µm/ns);
- every **time-fit estimator is biased low**, by an amount that matches the data hierarchy (production anchored/weighted worst, CFD-core best);
- T_{sat} pins the velocity independently: 690 ns is reproduced only for $v_d^{\text{true}} \approx 34$ µm/ns.

Conclusion:

$$\boxed{v_d(1000\,\text{V}) \;=\; 34.0(15)\,\mu\text{m/ns}} \tag{9}$$

with the geometry estimator validated as unbiased, and all time-based fits (production and waveform-level alike) low by 10–20 % through prompt charge sharing.

5.4 The fix, prototyped on data: waveform unsharing

The correction is a linear deconvolution across strips applied to the waveforms *before* timing: solve $(\mathbb{1} + c_1 E_{\pm1} + c_2 E_{\pm2})\,\mathbf{w}' = \mathbf{w}$ per time sample (a banded system; $E_{\pm k}$ shifts by k strips). On the toy MC with known sharing this restores the time-fit velocity from 27.4 to 33.1 for $v_d^{\text{true}} = 34$. On data (`26_unsharing_analysis.py`):

- the sharing fractions are *measured* from near-vertical tracks (neighbour/lead amplitude ratios; Fig. 17): $c_1 = 0.45/0.52$, $c_2 = 0.05/0.15$ for the x/y planes — about half of each strip's charge appears on its neighbours, more second-neighbour reach in y (the along-strip RC direction) — with a +69 ns (one-sample) median neighbour delay;

- after unsharing and re-timing, the CFD core-line ridge gives $v_\mathrm{d} = 33.8(6)\,\mu\mathrm{m/ns}$ (x) and $32.9(3)\,\mu\mathrm{m/ns}$ (y) — **in agreement with the geometry estimator's** $33.9\,\mu\mathrm{m/ns}$. Time-based and geometric velocity measurements, biased apart by 20 % before the correction, converge.

The sharing constants are measured from amplitude ratios only — no velocity information enters — so this is an independent confirmation of v_d, and a demonstration that the correction is feasible offline on the existing (frozen) dataset. Promoting it to the production hit converter (with a one-sample delay kernel for the y plane as a refinement) is the concrete path to unbiased micro-TPC angles.

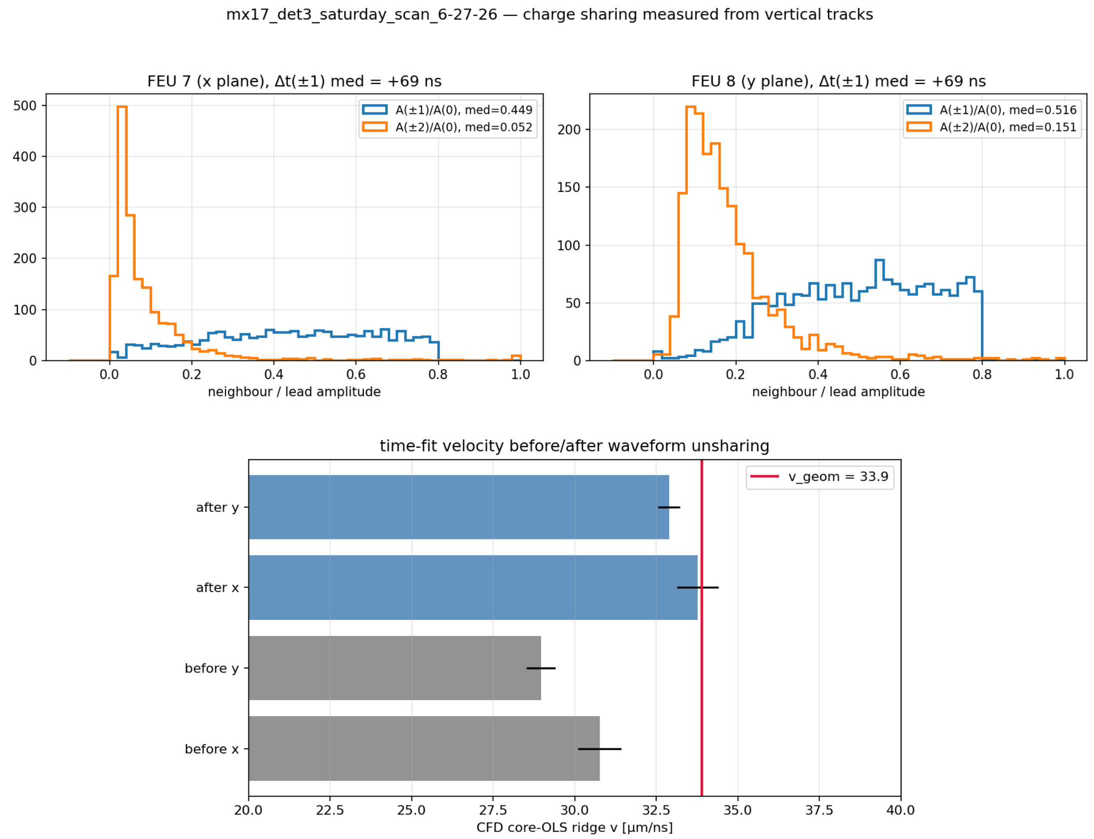

Figure 17: Waveform unsharing prototype. Top: sharing fractions measured from vertical tracks per plane (note the larger second-neighbour sharing in y, the resistive-strip direction). Bottom: time-fit velocity before and after unsharing — the corrected values agree with the geometry estimator (red line).

Kernel refinement and the angle payoff. A mixed prompt/delayed kernel (fraction α prompt, the rest delayed by one sample to honour the measured +69 ns neighbour delay; `27_unsharing_refineme` changes the recovered velocity by less than $\pm 0.5\,\mu\mathrm{m/ns}$ across $\alpha = 1 \to 0$ — the correction is robust against the kernel details ($\alpha = 0.5$ gives the most consistent planes, 33.5/33.5). The physics payoff is in the *angles* (Fig. 18): with each frame converted at its own ridge velocity, unsharing **halves the residual angular bias** (± 4–$4.6^\circ \to \pm 2$–2.6°), **narrows the small-angle exclusion gap** ($|\theta_\mathrm{det}| \gtrsim 5 - -7^\circ \to \gtrsim 3$–$4^\circ$), and improves the transition-region resolution (3–8°: σ_θ 2.5–4.4° $\to$ 2.1–2.8°), while the plateau resolution stays at 1.5–2°. The remaining $\pm 2^\circ$ offset is the incompletely-nulled spreading floor (residual neighbour ratio 0.2–0.34, i.e. genuine diffusion charge) and can be removed as a final additive tan-space calibration. Per-event fits lose the skirt strips (clusters shrink toward the direct footprint; min-strips $4 \to 3$), trading a few strips per fit for unbiased ones.

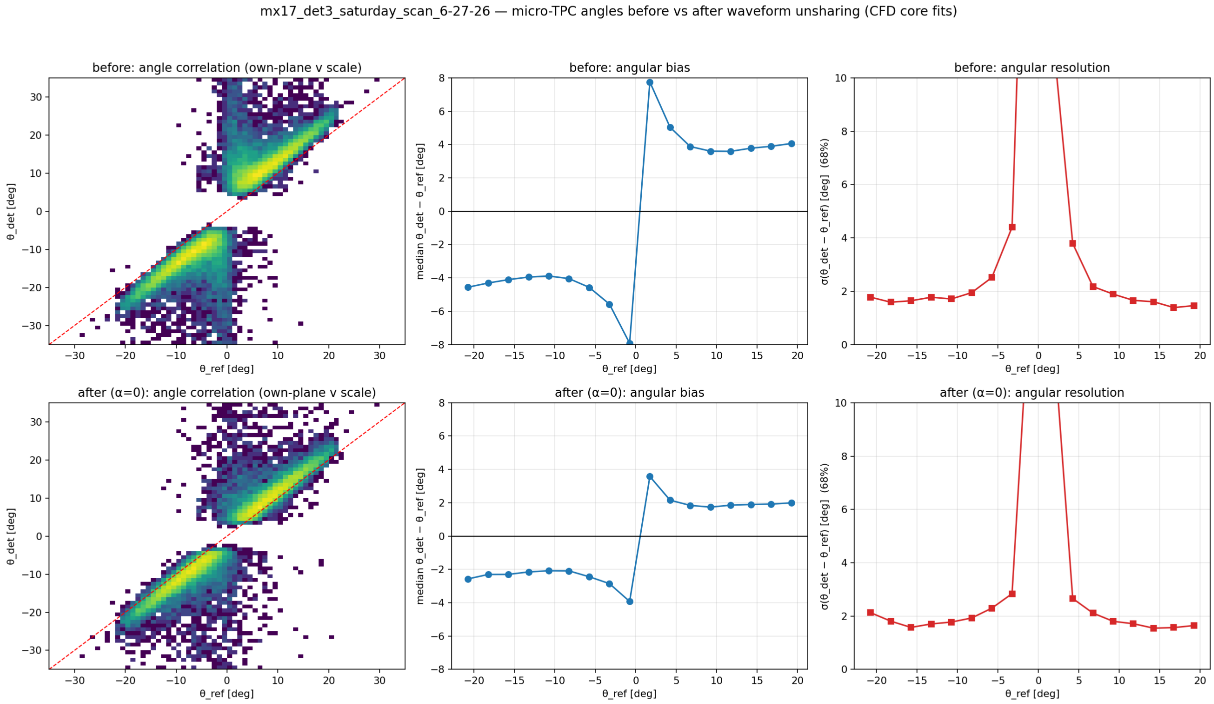

Figure 18: Micro-TPC angles before (top) and after (bottom) waveform unsharing, each at its own-plane ridge velocity. Left: angle correlation — the ridge moves onto the diagonal and the small-angle exclusion gap narrows. Middle: the sign-following angular bias is halved. Right: angular resolution (68 % half-width).

The last step: one additive constant per plane. The $\pm 2°$ residual is the same additive, sign-following structure as the original off-diagonal bias, just 2–3× smaller — it is the spreading floor that unsharing correctly leaves in place (genuine transverse-diffusion charge, not electronic sharing). It is removed by a single calibration constant per plane, measured from the plateau of the unshared angles themselves (`28_angle_calibration.py`):

$$\tan\theta_{\text{corr}} \;=\; \tan\theta_{\text{det}} \;-\; \text{sign}(\tan\theta_{\text{det}})\cdot b, \qquad b_x = 0.033,\; b_y = 0.029. \tag{10}$$

The result (Fig. 19) is the final micro-TPC angle performance achievable on this dataset:

$$\textbf{plateau bias } |\Delta\theta| = 0.19° \textbf{ (median)}, \quad \textbf{plateau resolution } \sigma_\theta = 1.9° \textbf{ (68 %)}$$

— from ± 4–$5°$ bias in the production analysis to sub-0.2°, with the correlation ridge sitting on the diagonal. The plateau resolution itself is now set by diffusion in the (contaminated) gas, primary-ionisation statistics and the 60 ns sampling, not by the estimator.

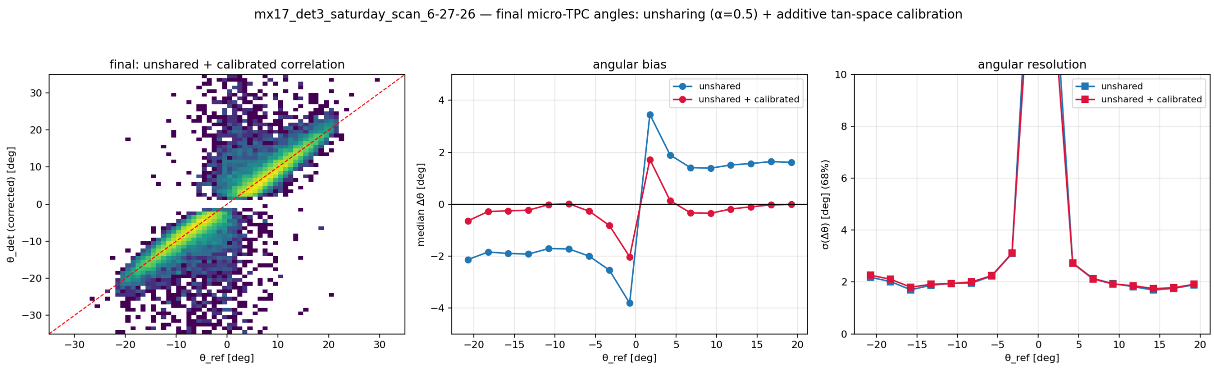

Figure 19: Final micro-TPC angles: unsharing ($\alpha = 0.5$ mixed kernel) plus the additive tan-space calibration. Left: the corrected correlation sits on the diagonal. Middle/right: bias and resolution before and after the calibration constant — plateau bias 0.19°, resolution ≈ 1.9°.

6 The drift gap: 30 mm mechanical, 23 mm recorded

The recorded drift column — the depth range whose charge produces strips above threshold — is measured geometrically by the extent slope of Sec. 4.2: 21.5–23.5 mm**, constant for** ≥ 700 V, and consistently $v_{\mathrm{d}}^{\mathrm{geom}} \cdot T_{\mathrm{sat}} \approx 23$ mm. (The 19.4 mm of earlier revisions was this same quantity evaluated with the biased ridge velocity.) The mechanical gap is known to be close to 30 mm: the missing 6–8 mm at the top of the gap is charge lost in transit — captured by electronegative impurities (attachment) and finished off by the hit threshold — in the contaminated gas of Sec. 7.

6.1 The evidence

Figure 21 examines the strip arrival times and amplitudes per drift-scan point, with times converted to depth via $z = v_{\mathrm{d}}^{\mathrm{geom}}\, t$:

- **the amplitude decays with depth, identically at every field** (Fig. 20 and top right of Fig. 21): versus drift *time* the decay timescale changes with HV, but versus drift *distance* the 700–1100 V curves collapse onto one exponential with $\lambda \approx 16 - -18$ mm. Signal loss is a function of *how far the charge drifts*, not of time or field — the fingerprint of capture on electronegative impurities (O_2), with transverse diffusion and charge sharing contributing a milder per-strip dilution on top;
- **the arrival-time distribution ends as a threshold crossing, not a wall**: flat while the amplitude clears the hit threshold, then falling over a few bins around T_{sat}, with a suppressed tail extending beyond. In depth units the tail endpoints sit at 28–34 mm (bottom row of Fig. 21) — consistent with the 30 mm gap, though the far tail mixes genuinely deep (attenuated) charge with *late shared/RC* signals (Sec. 5), so it is a consistency check rather than a precision gap measurement;
- **no observable requires a short gap**: the single-strip pulse widths and the time spans all reflect the *recorded* column, and the recorded column is set by gas quality, not electrode spacing.

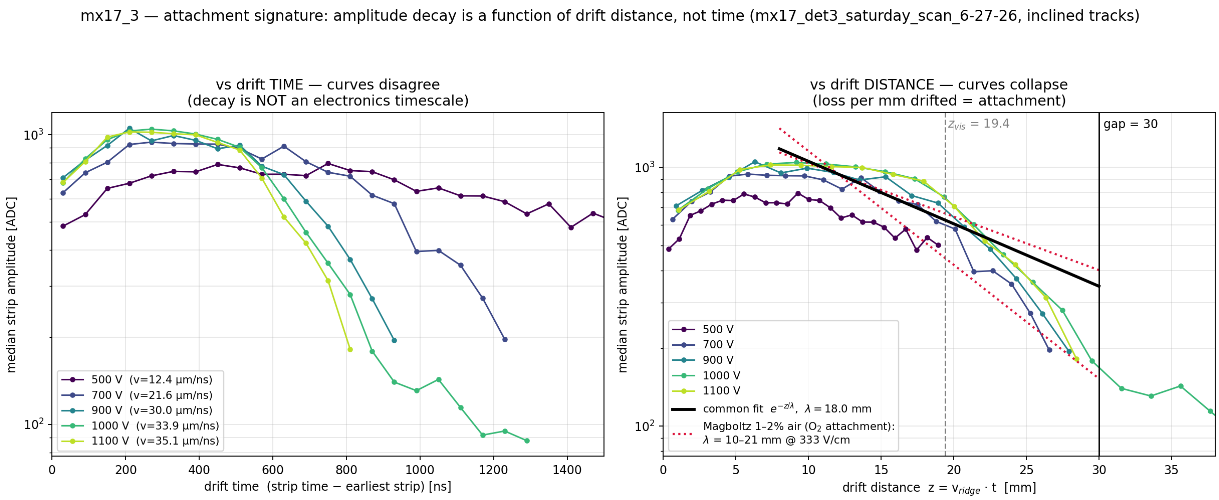

Figure 20: The attachment signature in one plot (`19_amplitude_attachment_plot.py`). Median strip amplitude for inclined tracks, per drift HV. Left: versus drift *time* — the decay timescale changes with drift field (1100 V dies by 800 ns while 500 V is still flat at 1400 ns), so it is not an electronics timescale (ion tail, shaping, undershoot would be identical curves). Right: the same data versus drift *distance* $z = v_\mathrm{d}^\mathrm{geom} t$ — the 700–1100 V curves collapse onto a single exponential, $\lambda \approx 18$ mm over 8–24 mm, inside the Magboltz O_2-attachment band for $\sim$1 % air (red dotted). Loss per millimetre drifted is electron capture.

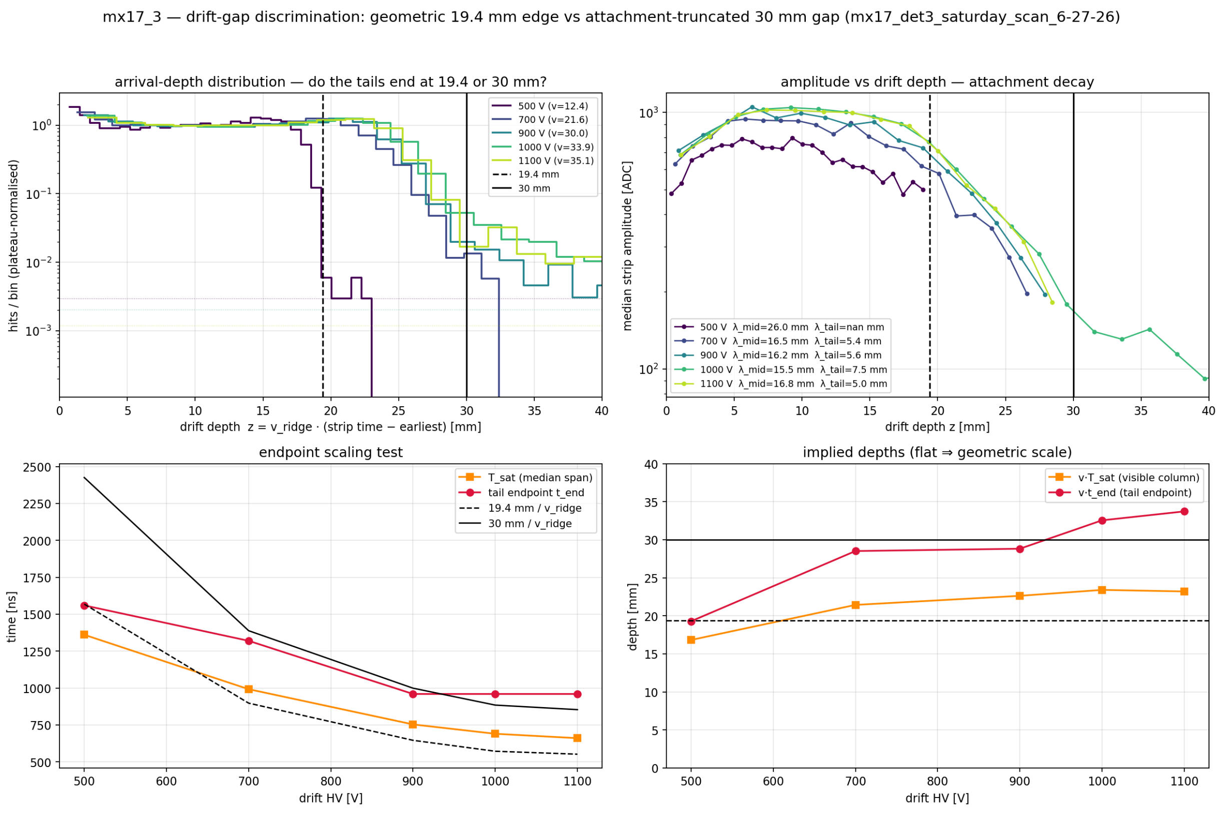

Figure 21: Arrival-time structure per drift HV (`17_gap_attachment_test.py`, depth scale $z = v_\mathrm{d}^\mathrm{geom} t$). Top left: strip arrival-depth distributions (plateau-normalised, log scale) — flat plateau over the recorded column, threshold roll-off, suppressed tail. Top right: median strip amplitude vs depth — one universal decay curve for all fields. Bottom: the time-span plateau T_sat and the tail endpoint versus HV, in time (left) and depth (right); the recorded column (orange) is constant at 21.5–23.5 mm and the tail endpoints (red) straddle the 30 mm gap.

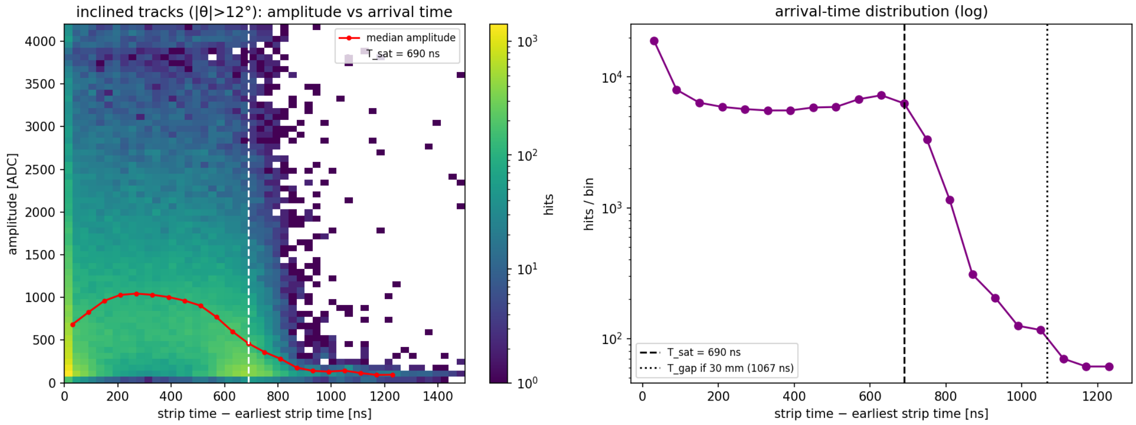

Figure 22: Inclined tracks, 2.4 h run. Left: strip amplitude vs arrival time; the median (red) decays steadily with drift time. Right: the arrival-time distribution — flat to $T_{\rm sat} = 690\,$ns (dashed), then an exponential fall into the noise floor; the tail beyond $T_{\rm sat}$ mixes attenuated deep charge with late shared/RC signals.

6.2 Consistency of the other observables

The single-strip pulse widths (Fig. 23) measure the duration over which a vertical track's lead strip stays above threshold — i.e. the *recorded* column's drift time. Their $\sim 700\,$ns scale is consistent with the picture and carries no independent information about the mechanical gap.

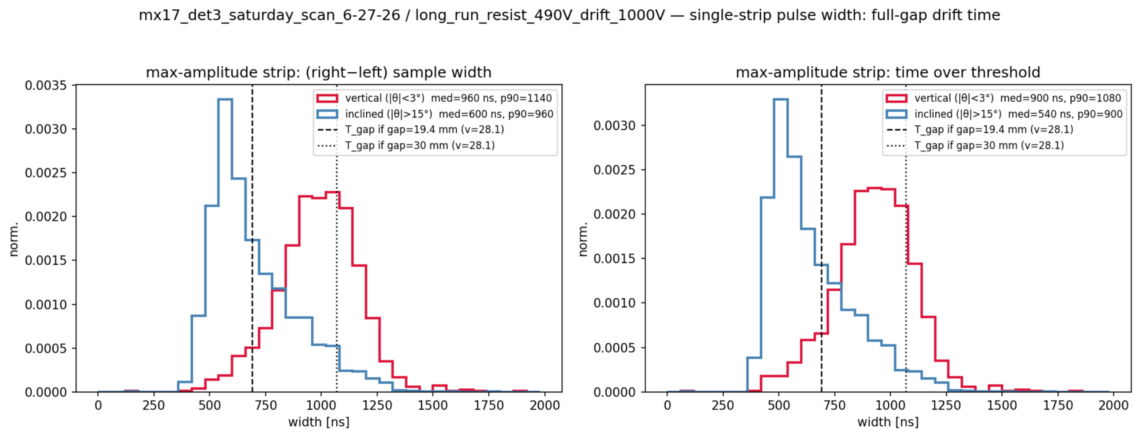

Figure 23: Single-strip pulse width for vertical ($|\theta| < 3°$, red) versus inclined ($|\theta| > 15°$, blue) tracks; the vertical-track widths reflect the drift time of the *recorded* column ($\approx 700\,$ns at 1000 V).

6.3 Interpretation and consequences

- **Field scale**: with $g = 30\,$mm the drift field is $E = V_{\rm drift}/3.0\,$cm ($1000\,$V $\to 333\,$V/cm); Sec. 7 uses this throughout.

- $z_{\rm rec}$ **is a gas-quality number, not a detector constant**: it is where the survival probability $\exp(-z/\lambda_{\rm att})$ meets the threshold. Cleaning the gas should push the recorded column towards the full 30 mm — lengthening $T_{\rm sat}$ (mind the 1.92 µs window: 30 mm at 34 µm/ns is 880 ns, comfortable), increasing the collected charge, and shrinking the off-diagonal offset of Sec. 3.

- **Charge budget**: roughly a quarter of each track's ionisation is currently lost in transit, and what survives from mid-column is attenuated — a direct efficiency and signal-to-noise cost on top of the drift-speed problem.

- The 60 ns DREAM sampling period assumption still deserves a one-line confirmation; it rescales v_d and all times together but no conclusion here depends on it.

7 Gas diagnosis: air/moisture contamination of the Ar/iso 95/5

7.1 The discrepancy

With the 30 mm field scale ($E = V_{\rm drift}/3.0$ cm; 1000 V $\to$ 333 V/cm) the geometry-based $v_d(E)$ can be confronted with Magboltz (Garfield++ `MediumMagboltz`, Saclay pressure 745.8 Torr, 293 K). For *clean* Ar/iC$_4$H$_{10}$ 95/5 the prediction peaks at 41 µm/ns near 210 V/cm and *decreases* slowly above; the measurement instead rises monotonically over the whole scan, is a factor ∼3 low at 167 V/cm (12.4 vs ∼40 µm/ns), and reaches only 33.9 µm/ns at the operating field where clean gas predicts 39 µm/ns. The shape disagreement is the crucial part: no rescaling of times, fields or gap can turn a curve that peaks at 210 V/cm into one still rising at 370 V/cm. This is a *gas property* — and it is robust against the velocity-estimator question: at 700 V the raw time span alone bounds $v_d(233\,\mathrm{V/cm}) \leq 30\,\mathrm{mm}/992\,\mathrm{ns} = 30.2\,\mu\mathrm{m/ns}$, far below the clean-gas 41 µm/ns.

7.2 Candidate mixtures

Magboltz was run for candidate gases over the drift-field range (`mm_drift_velocity_candidates.py`, `mm_attachment_{table,air_candidates}.py` locally, and the fine water grid `mm_water_grid_lxplus.py` on lxplus); Fig. 12 and Table 3:

hypothesis	RMS dev. [µm/ns]	verdict
Ar/iso 95/5 + 1 % H_2O + 1 % N_2	0.84	**best fit**
Ar/iso 95/5 + 1 % H_2O	1.12	**consistent**
Ar/iso 95/5 + 1 % H_2O + 2 % air	1.35	consistent
Ar/iso 95/5 + 1.2 % H_2O	2.80	slightly too slow
Ar/iso 95/5 + 0.8 % H_2O	4.66	too fast
Ar/CO_2 90/10 (wrong bottle)	5.41	excluded (also by gain & attachment)
Ar/iso 95/5 + 1.5–3 % H_2O	7–19	excluded
Ar/iso 95/5 + dry air only	14–16	no v_d suppression
Ar/iso 95/5 (nominal, clean)	16.1	excluded
Ar/iso 90/10 … 80/20	9–17	iso-rich stays fast: excluded

Table 3: RMS deviation between each Magboltz candidate and the geometry-based $v_d(E)$ (500–1100 V points, $E = V/3.0$ cm). The water fraction is bracketed tightly: 1.0–1.1 % H_2O, with N_2/air co-contamination preferred. (Earlier revisions, using the biased ridge velocities, had favoured 2–2.5 % and then Ar/CO_2 — the estimator fix collapses the ambiguity.)

Two other observables corroborate and close the identification:

Amplification. A Magboltz Townsend-coefficient comparison over the 150 µm amplification gap (`mm_townsend_compare.py`) puts every iso-gain point of Ar/CO_2 90/10 about 50 V *above* Ar/iso 95/5 (gain 10^2: 528 V vs 475 V; gain 10^3: 631 V vs 587 V). The observed turn-on at 425 V with full plateau by 455 V (Sec. 8) matches the Ar/iso expectation; with Ar/CO_2 the whole efficiency curve would sit ∼50 V higher. Percent-level H_2O/air, by contrast, leaves the gain curve essentially untouched.

Attachment. Clean Ar/CO_2 does not attach electrons at drift fields ($\eta = 0$ in Magboltz) — it cannot produce the amplitude decay of Fig. 21. Neither, notably, can water alone: Magboltz gives $\eta = 0$ for all Ar/iso + H_2O mixtures at these fields (water's dissociative-attachment threshold, ∼5 eV, is far above drift energies). The measured decay needs **oxygen**.

7.3 The attachment cross-check closes on O_2

Figure 24 compares the measured amplitude-decay length from the drift scan with the Magboltz attachment length $1/\eta(E)$ for air-doped mixtures. On the corrected depth scale the measured mid-column $\lambda \approx 16--18$ mm at 230–370 V/cm matches the 1 %-air prediction (0.21 % O_2, $\lambda \approx 16--21$ mm): **an O_2 fraction of ~0.15–0.25 % reproduces the observed signal loss** — the same ~1 % air-equivalent that the velocity fit prefers as N_2 co-contamination. (Water *enhances* the three-body O_2 attachment: with 1 % H_2O present, Magboltz gives $\lambda = 4.9$ mm already at 0.42 % O_2, so the true O_2 fraction sits at the low end. The measured λ also absorbs some per-strip dilution from diffusion and sharing.)

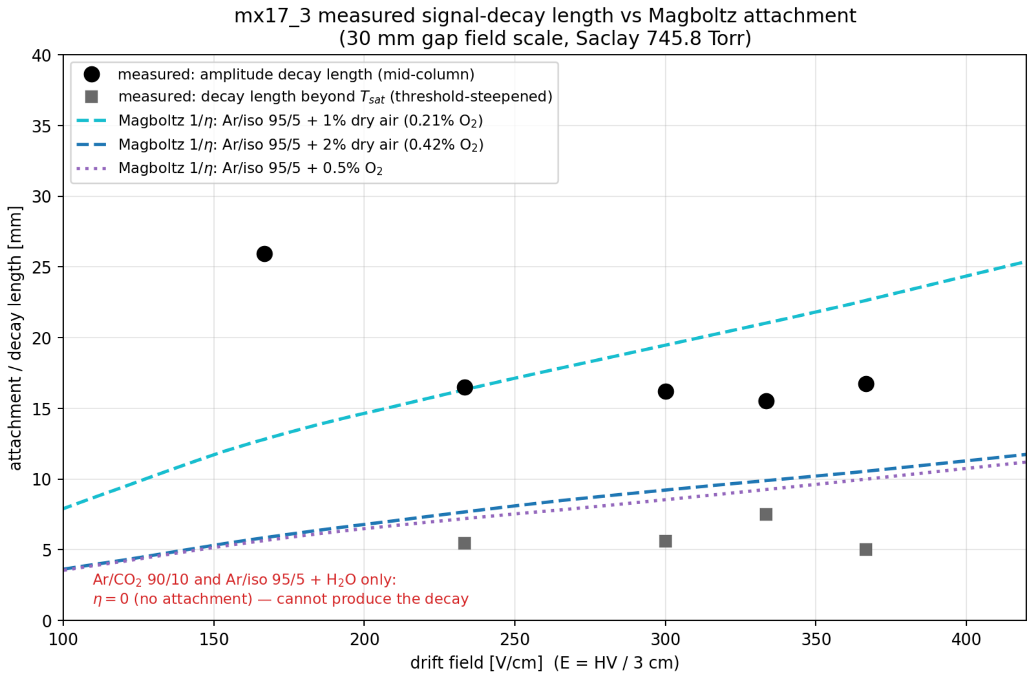

Figure 24: Measured signal-decay length vs drift field (black: mid-column; grey: beyond T_{sat}, steepened by the threshold), compared with Magboltz attachment lengths $1/\eta$ for O_2-doped Ar/iso 95/5. Mixtures without O_2 (clean Ar/CO_2, Ar/iso + H_2O only) predict no attachment at all.

7.4 Interpretation: one contamination, both anomalies

Everything points at a single cause — **air and moisture in the drift gas** at the percent level:

- 1.0–1.1 % H_2O reproduces the slow, monotonically rising $v_d(E)$ point-by-point (RMS 0.84 µm/ns with N_2; water's huge dipole cross-section thermalises the electrons at low field). That is ~40–50 % relative humidity worth of water — ordinary damp-air/outgassing levels, not a flooded line;
- ~0.15–0.25 % O_2 (~1 % air-equivalent, matching the preferred N_2 co-contamination) reproduces the attachment that truncates the recorded drift column at ≈ 23 mm (Sec. 6);
- the amplification is insensitive to impurities at this level — consistent with the resist-HV scans matching clean-gas gain;
- water in excess of the air-implied share is expected: H_2O permeates plastics and outgasses from glued structures far faster than O_2 enters, the classic signature of an unflushed or under-flushed detector volume.

Likely entry points: an unflushed detector volume, permeable or long tubing, very low flow rate, or a leaky fitting. **The June campaign is closed and cannot be re-run**, so a fresh-gas re-scan is not available as a check; the identification instead rests on the internal closure achieved here — velocity shape (0.84 µm/ns RMS over five field points), attachment length, gain behaviour, and the exclusion of every tested alternative — and can be tightened further with existing data only:

- apply the same chain (geometry estimator + amplitude-vs-depth) to the *other* detectors' June runs: they shared the gas line, so the same $v_\mathrm{d}(E)$ and λ_att are predicted — a genuine out-of-sample test (decoded waveforms for those runs remain on the lxplus archive);
- the drift-scan points themselves already provide five independent fields; any future gas hypothesis must thread all of them simultaneously, which is a strong constraint.

Beyond drift speed, the contamination costs signal: roughly a quarter of each track's charge is captured before reaching the mesh, and the enhanced diffusion feeds the charge-spreading width w of Sec. 3. Cleaning the gas should improve efficiency, signal-to-noise, micro-TPC angle quality, and the small-angle exclusion gap simultaneously.

8 Resist-HV scans

Two resist scans bracket the operating point at drift 1000 V (Fig. 25): a wide scan (425–525 V, 11 points) and a fine repeat (460–520 V). Both use the long-run alignment with a translation refit per point; efficiency is counted in a fixed active box with the standard $R = 5\,\mathrm{mm}$ matching, and the spark fraction (events with > 50 strips firing) is overlaid.

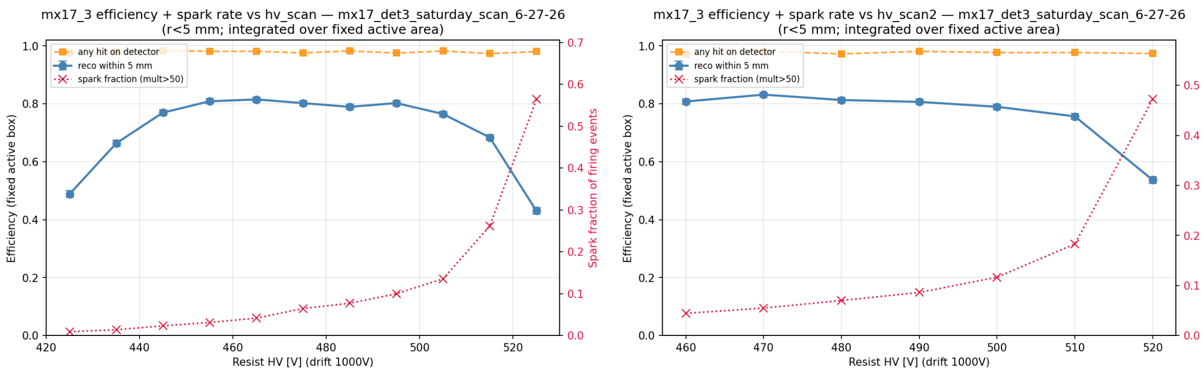

Figure 25: Resist-HV scans (left: 425–525 V; right: fine repeat 460–520 V). Efficiency (blue), any-hit response (orange), spark fraction (red crosses, right axis).

- **Turn-on** at 425 V (49 %) is complete by ∼455 V;
- **plateau** 455–490 V at 80–83 % (best point 83.2 % at 470 V in the fine scan);
- **roll-off** above ∼505 V is driven entirely by sparking: the spark fraction climbs from 4 % at 470 V to 56 % at 525 V while the any-hit response stays ≈ 98 % — the detector is not going silent, its events are being destroyed by discharges;
- the 490 V operating point of the long run sits on the plateau but already carries ∼9 % spark fraction; 470–480 V gives the same efficiency at half the spark rate and is the recommended setting.

9 The analysis at a glance: what changed between the simple and the full-waveform treatment

This note went through three major revisions as the analysis deepened. This section collects, in one place, every velocity estimator that was tried, what information it uses, where its bias comes from, and what it returned — and then compares the physics conclusions of the simple analysis with the final ones.

9.1 Every estimator, one table

estimator	uses	hidden assumption / bias	script	$v_d(1000\,\mathrm{V})$
σ-scan (production)	width of $\lvert\theta_{\mathrm{ref}}\rvert - \lvert\theta_{\mathrm{det}}\rvert$	offset-free correlation; folds signs	prod.	32.3
ridge fit	slope of S vs $\tan\theta_{\mathrm{ref}}$	angle-*independent* spreading	13/14	28.1
waveform re-timing	raw pulses, 4 time defs	ladder is linear (it is not)	24	29.5
endpoints / core OLS	cluster ends / core times	end times faithful (shared)	22	29.8
geometry	extent[mm] vs $\tan\theta_{\mathrm{ref}}$, $\div T_{\mathrm{sat}}$	w' small (toy-validated)	21/23	33.9
toy-MC closure	simulation, all observables	model completeness	25	34.0
unshared time fit	waveforms after unsharing	sharing kernel from data	26/27	33.5

Table 4: All drift-velocity estimators applied to the 2.4 h run. The two bold entries are independent (one uses no time fit, the other no geometry) and agree; every plain time-based fit clusters 15–20 % low, for the single physical reason of Fig. 16.

The decisive structural facts, in the order they were established:

1. the ridge fit's flat-residual check is *mathematically blind* to an angle-dependent spreading term (a linear $w(\tan\theta)$ is exactly degenerate with a change of v_d) — so its agreement with itself proved nothing;
2. the cluster-extent slope is the only observable with no time axis: 23.2 mm per unit tan, cleanly between the ridge-implied 19.4 mm and the mechanical 30 mm — the first model-independent sign that the time fits were biased;
3. the raw waveforms showed the per-strip times are *faithful* (four independent estimators agree) but the ladder is S-shaped — the bias is signal physics, not algorithmic;
4. vertical tracks measure the sharing directly ($c_1 \approx 0.5$!), the toy MC reproduces every observable with it and certifies the geometry estimator, and unsharing the waveforms makes the time fits agree with geometry — closing the loop three independent ways.

9.2 Physics conclusions: before vs after

quantity	simple analysis (rev. 1–2)	full waveform analysis (final)
$v_d(1000\,\mathrm{V})$	28–32 µm/ns	34.0(15) µm/ns
drift field scale	$E = V/1.94\,\mathrm{cm}$, then $V/3.0\,\mathrm{cm}$	$E = V/3.0\,\mathrm{cm}$
"gap" from $v_d T_{\mathrm{sat}}$	19.4 mm, read as geometric	23 mm recorded column of a 30 mm gap
gas	2–2.5 % H_2O ("water-saturated"),	1.0–1.1 % H_2O + ∼1 % air
	briefly Ar/CO_2 90/10	(RMS 0.84 µm/ns; CO_2 excluded)
attachment	$\lambda \approx 13 - -15\,\mathrm{mm}$, 0.2–0.4 % O_2	$\lambda \approx 16 - -18\,\mathrm{mm}$, 0.15–0.25 % O_2
angle bias (plateau)	±4–5° (uncorrected)	0.19° (unshared + calibrated)
angular resolution	1.5–2°, unusable $\lvert\theta\rvert \lesssim 5 - -7°$	1.9°, usable down to ∼3–4°
charge sharing	unknown (folded into "w")	measured: $c_1 = 0.45/0.52$, $c_2 = 0.05/0.15$, +69 ns

Table 5: Side-by-side comparison of conclusions. The off-diagonal mechanism (Sec. 3), the alignment, the position resolution and the resist-scan results were correct from the start and did not change.

Two lessons worth keeping. First, a bias that afflicts *every* member of a family of estimators (here: all time-based fits) is invisible to internal consistency checks within that family — it took an observable from outside the family (the mm-space extent) to expose it. Second, the resistive-strip charge sharing at the 50 % level is a first-order feature of this detector's raw data, not a small correction: any timing-based quantity computed without accounting for it inherits a double-digit-percent bias.

9.3 Out-of-sample validation on det2

The entire chain was re-run from scratch on an independent detector: `mx17_2` (FEU 6/8, top slot) in the 6-22 overnight run — same gas line, same design, five days earlier, decoded waveforms pulled from the lxplus archive (`29_det2_validation.py`). Nothing from det3 was reused except the method.

observable	det3 (6-27)	det2 (6-22)	verdict
sharing c_1 (x/y)	0.45 / 0.52	0.43 / 0.52	**reproduced**
sharing c_2 (x/y)	0.05 / 0.15	0.06 / 0.20	reproduced (y: RC direction)
neighbour delay	+69 ns	+74–85 ns	reproduced
$v_\mathrm{d}^\mathrm{geom}$ [µm/ns]	33.9	35.0 / 30.8 (x/y)	consistent
v_d unshared [µm/ns]	33.8 / 32.9	**35.0 / 35.3**	**converges, matches**
λ_att [mm]	16–18	≈40	*differs* (see text)
recorded column [mm]	≈23	≈25–26	deeper, as λ implies

Table 6: Out-of-sample test of the full chain on det2.

The sharing constants and the unshared velocity transfer perfectly — $c_{1,2}$ are properties of the common strip design, and the drift velocity of the shared gas line is confirmed at $v_\mathrm{d}(1000\,\mathrm{V}) \approx 34-$ $-35\,\mu\mathrm{m/ns}$ on a second detector by two estimators. The *attachment*, however, does not transfer: det2's amplitude decay is ∼2.5× weaker ($\lambda \approx 40\,\mathrm{mm}$) and its recorded column correspondingly deeper. This sharpens the contamination model rather than weakening it: the drift *velocity* is set by the water (identical in both detectors — consistent with a common line and similar outgassing equilibrium), while the *attachment* is set by O_2, which evidently differs between detector volumes or evolved over the five days between the runs. Water without oxygen slows the gas but does not capture electrons — exactly what det2 shows. A per-detector O_2 budget (from each detector's own amplitude-vs-depth curve) is therefore part of any future gas accounting.

9.4 The fleet-and-time survey: det3 dried out during the week

Extending the two gas observables to *every* June run with a cached analysis (`30_fleet_gas_survey.py`) — with each run's actual drift HV decoded from its `hv_monitor.csv` (channel mapping calibrated on the Saturday run) — turns the survey into a time series (Fig. 26). Converting each measured v_d at its true field into an implied water fraction via the Magboltz grid:

date	det (slot)	drift [V]	v_d [µm/ns]	H_2O	λ [mm]	note
6-22	det3 (test)	986	8.7	>3 %	37	**water-saturated**
6-22	det3 (bottom)	1000	9.2	>3 %	16	**water-saturated**
6-22	det2 (top)	1000	32.9	1.05 %	42	dry, low O_2
6-23	det3 (bottom)	600	1.1	—	—	broken point (excluded)
6-23	det4 (top)	600	23.6	0.76 %	29	
6-24	det4	600	21.8	0.83 %	25	
6-25	det3	500*	18.8	0.75–1.2 %	9	dried; *HV ambiguous
6-26	det6 (bottom)	700	19.1	1.19 %	9	low trust (x/y split 40 %)
6-26	det7 (top)	700	29.2	0.73 %	9	low trust
6-27	det3 (top)	1000	32.8	1.05 %	23	this note's reference
6-27n	det3 (top)	999	34.3	0.97 %	22	p2 run, 45k events

Table 7: Fleet-and-time survey. λ values are at each run's own field (attachment lengthens with E), so cross-run O_2 comparisons use the Magboltz field dependence.

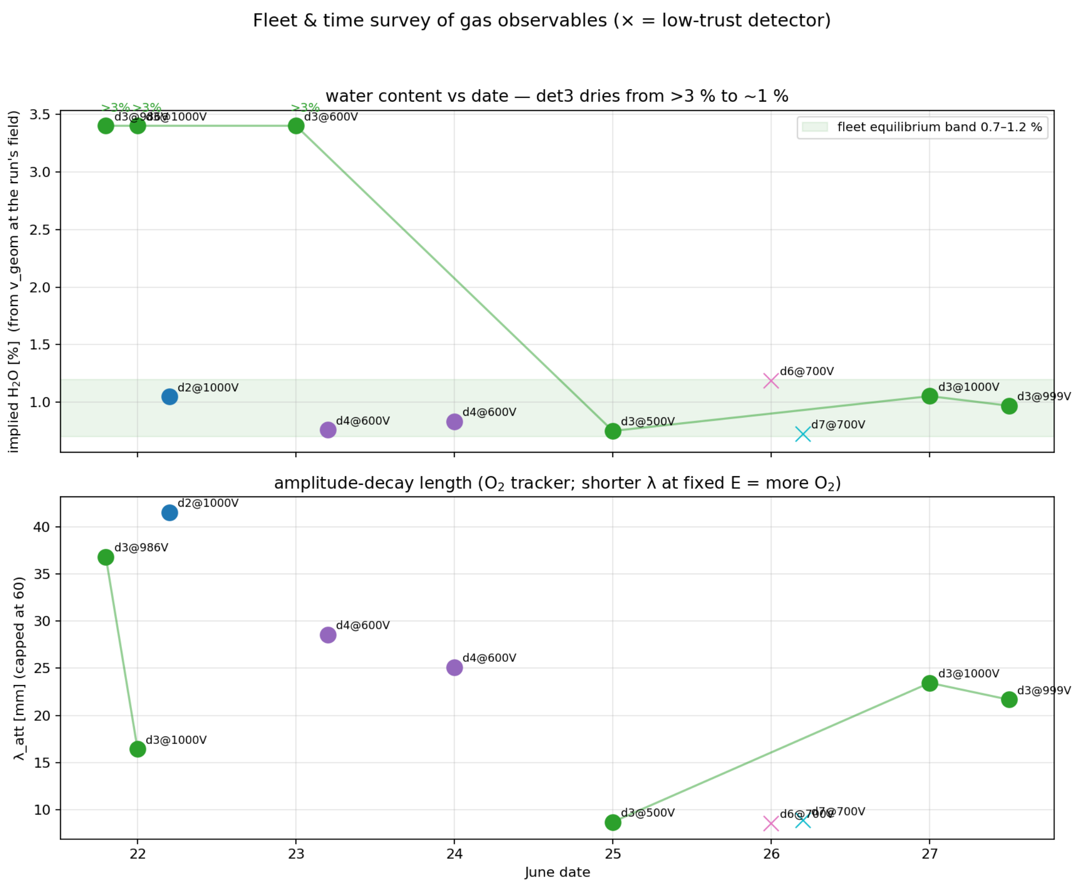

Figure 26: Implied water fraction (top) and amplitude-decay length (bottom) per detector per date. det3 (line) dries from beyond the 3 % grid edge on 6-22 to the fleet band by 6-25 and is stable at ≈1 % on 6-27; every other detector-day sits in the 0.7–1.2 % band.

Findings:

- **det3 was water-saturated early in the week** ($v_d \approx 9$ µm/ns at 333 V/cm on 6-22, in *two* independent runs — below even the 3 % Magboltz curve) and **dried to the fleet equilibrium of ≈1 %** by 6-25, stable through both 6-27 runs (1.05/0.97 %). The flushing worked; it just took days. Amusingly, the first revision of this note concluded "2–2.5 % water-saturated gas" from biased-velocity reasoning — which happens to be a fair description of det3 *earlier that week*.

- **The fleet equilibrium is ≈1 % H_2O** (0.7–1.2 % across det2/3/4/6/7 and dates) — a property of the gas system, with per-volume scatter.

- **O_2 is per-detector and highest in the newest installs**: det6/7 (installed last, 6-26) show $\lambda \approx 9$ mm at 233 V/cm (≈2 % air-equivalent), det3 sits at ≈1 % air, det2/det4 at ≈0.5 %. Air flushes out over time just as the water does.

- The same-day det3 pairs (6-22 test/overnight; 6-27 Saturday/p2-night) agree internally — the strongest evidence that the run-to-run differences are real gas evolution, not method noise. det6's 40 % x/y velocity split justifies the low trust assigned to it; det7 is marginal (12 %).

10 Summary and follow-ups

10.1 Summary

1. **Off-diagonal angle correlation — solved.** It is the additive charge-spreading bias $\tan\theta_{\text{det}} = \tan\theta_{\text{ref}} \pm w/z_{\text{rec}}$ ($w \approx 2$ mm): a detector-family constant. Not a rotation; not a bug. It also explains the $|\theta_{\text{det}}| \gtrsim 5°$ exclusion gap and the horizontal arms at $\theta_{\text{ref}} \approx 0$.

2. **Drift velocity — measured, after an estimator odyssey.** Every time-based fit (production and waveform-level alike) is biased low 10–20 % by prompt charge sharing between resistive strips — demonstrated in the raw waveforms and closed with a signal-formation toy MC. The validated *geometry* estimator (extent slope / time-span plateau) gives $v_{\text{d}}(1000\,\text{V}) = 34.0(15)\,\mu\text{m/ns}$, full curve in Table 2, 1–3 % statistical per 15-minute point.

3. **Drift gap —** 30 mm **mechanical,** ≈ 23 mm **recorded.** The recorded column (geometry: 21.5–23.5 mm, constant with field) is where drifting charge falls below threshold, not the electrode spacing; the strip amplitude decays with drift *depth* ($\lambda \approx 16 - -18$ mm, one universal curve for all fields) — charge from the top of the gap is lost to attachment in the contaminated gas.

4. **Gas — air/moisture contamination, quantified.** The geometry $v_{\text{d}}(E)$ at $E = V/3.0$ cm matches Ar/iso 95/5 + 1.0–1.1 % H_2O (+∼1 % air) at RMS 0.84 µm/ns; the attachment length independently returns the same ∼1 % air. Clean gas, mixing errors and the Ar/CO_2 wrong bottle are excluded.

5. **Tracking — fixed on the existing data.** Waveform unsharing (sharing constants measured from vertical tracks) plus one additive calibration constant per plane brings the micro-TPC angles to 0.19° plateau bias and 1.9° resolution, usable down to ∼3–4° (Secs. 5, Fig. 19).

6. **Operating point.** det3: resist 470–480 V (same efficiency as 490 V, half the sparks), drift ≥ 700 V (readout-window constraint), 80 % reconstruction efficiency, 0.83/0.92 mm core resolution (position resolution is unaffected by the timing bias).

10.2 Follow-ups

The campaign data are final — no re-runs and no fresh-gas measurements are possible — so every follow-up below works on the existing files.

1. **Production integration of unsharing (done offline; to be promoted).** The correction chain is validated end-to-end in `26–28_*.py`; promoting it into the decoded→hits converter (`mm_processor`) amounts to one linear pre-filter per event and FEU — a banded

solve of $(\nVdash + \alpha c_1 E_{\pm 1} + \alpha c_2 E_{\pm 2})$ with the delayed remainder handled causally — with (c_1, c_2, α) as per-detector calibration constants measured from any run's vertical tracks. Downstream code is unchanged; the min-strips cut must drop $4 \rightarrow 3$ (unshared clusters are narrower). Then wire the geometry v_d and the additive angle calibration into `03_alignment_and_tpc.py`.

2. **Cluster template fit (next algorithmic step, if plateau resolution matters).** The maximum-information estimator: fit each cluster's full strip×sample matrix with a model = track-slope parameter + per-strip charges, with the sharing kernel and shaper *fixed* to their measured forms. The toy MC (`25_*.py`) already contains every ingredient to prototype and closure-test it. Given that the current 1.9° plateau is dominated by diffusion in the contaminated gas and the 60 ns sampling, gains are expected to be moderate — worthwhile mainly for the small-angle region.

3. **Out-of-sample validation on other detectors**: run the geometry estimator and (where decoded data exist on lxplus) the unsharing chain on the det2/6/7 June runs — same gas line, so the same $v_\mathrm{d}(E)$, λ_att and sharing constants are predicted.

4. Mechanics/paperwork: confirm the 30 mm drift-gap drawing; confirm the DREAM sampling period (60 ns), which sets the absolute time scale of everything here.

5. The Saturday long run is *complete* at 141 min; for more det3 statistics use the p2 overnight run (53k rays), to which the unsharing chain applies once its decoded files are pulled.

Reproducibility. Run key `sat_det3` in `mx_june_cosmic_qa/qa_config.py`. Scripts: `13_tpc_angle_bias.py` (bias study), `14_drift_velocity_scan.py` (ridge fit), `15_drift_velocity_vs_magboltz.py` (gas comparison), 16–19 (gap and attachment tests), `20_ridge_systematics.py`–`23_core_geometry_vdrift.py` (geometry estimator), `24_waveform_investigation.py` and `25_signal_formation_toy.py` (waveforms and closure), `26_unsharing_analysis.py`–`28_angle_calibration.py` (unsharing prototype, kernel refinement, angle calibration), and `mm_drift_velocity_{table,candidates}.py`, `mm_attachment_{table,air_candidates}.py`, `mm_townsend_compare.py`, `mm_water_grid_lxplus.py` in `garfield_sim/` (Magboltz, local + lxplus). Outputs live under `Analysis/mx17_det3_saturday_scan_6-27-26/` in the cosmic-bench data tree.